



Date: 25/06/26

Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:

- i. ipconfig
- ii. ping
- iii. getmac
- iv. systeminfo
- v. traceroute / tracert
- vi. netstat
- vii. nslookup
- viii. hostname
- ix. pathping x.arp

1. ipconfig

Description:

----- Details About Command -----

No.	Option	Description
1	ipconfig /all	Displays complete TCP/IP configuration information for all network adapters.
2	ipconfig /release	Releases the current IPv4 address assigned by the DHCP server.
3	ipconfig /renew	Requests a new IPv4 address from the DHCP server.
4	ipconfig /flushdns	Clears the DNS resolver cache stored on the computer.
5	ipconfig /displaydns	Displays the contents of the DNS resolver cache.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



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Date: 25/06/26

```
D:\>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 3 while it has its media disconnected.
An error occurred while releasing interface VMware Network Adapter VMnet1 : An address has not yet been associated with the network endpoint.

An error occurred while releasing interface VMware Network Adapter VMnet8 : An address has not yet been associated with the network endpoint.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . : fe80::910:a657:7d1f:ac96%15
    Autoconfiguration IPv4 Address. . : 169.254.246.15
    Subnet Mask . . . . . : 255.255.0.0
```

```
D:\>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-R2FNR93
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : lan

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : E4-E7-49-68-0D-1D
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
    Physical Address. . . . . : 82-2B-F9-2D-D3-C5
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
```

```
D:\>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```



Date: 25/06/26

```
D:\>ipconfig /displaydns

Windows IP Configuration

    1.160.30.172.in-addr.arpa
    -----
    Record Name . . . . . : 1.160.30.172.in-addr.arpa.
    Record Type . . . . . : 12
    Time To Live . . . . . : 432500
    Data Length . . . . . : 8
    Section . . . . . : Answer
    PTR Record . . . . . : DESKTOP-R2FNR93.mshome.net

    1.0.0.127.in-addr.arpa
    -----
    Record Name . . . . . : 1.0.0.127.in-addr.arpa.
    Record Type . . . . . : 12
    Time To Live . . . . . : 432500
    Data Length . . . . . : 8
    Section . . . . . : Answer
    PTR Record . . . . . : kubernetes.docker.internal
```

2. ping

Description:

----- Details About Command -----

No.	Option	Description
1	ping -t google.com	Continuously pings the destination until stopped manually (Ctrl + C).
2	ping -n 5 google.com	Sends a specific number of echo requests (here, 5 packets).
3	ping -l 1000 google.com	Sends packets with a specified size (here, 1000 bytes).
4	ping -w 5000 google.com	Sets the timeout to wait for each reply (here, 5000 milliseconds = 5 seconds).
5	ping -4 google.com	Forces the ping command to use the IPv4 protocol.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



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(2305CS332)

Date: 25/06/26

```
D:\>ping -n 5 google.com
```

```
Pinging google.com [2404:6800:4002:828::200e] with 32 bytes of data:  
Reply from 2404:6800:4002:828::200e: time=44ms  
Reply from 2404:6800:4002:828::200e: time=35ms  
Reply from 2404:6800:4002:828::200e: time=35ms  
Reply from 2404:6800:4002:828::200e: time=41ms  
Reply from 2404:6800:4002:828::200e: time=37ms  
  
Ping statistics for 2404:6800:4002:828::200e:  
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 35ms, Maximum = 44ms, Average = 38ms
```

```
D:\>ping -l 1000 google.com
```

```
Pinging google.com [2404:6800:4002:828::200e] with 1000 bytes of data:  
Reply from 2404:6800:4002:828::200e: time=39ms  
Reply from 2404:6800:4002:828::200e: time=45ms  
Reply from 2404:6800:4002:828::200e: time=157ms  
Request timed out.  
  
Ping statistics for 2404:6800:4002:828::200e:  
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 39ms, Maximum = 157ms, Average = 80ms
```

```
D:\>ping -t google.com
```

```
Pinging google.com [2404:6800:4002:828::200e] with 32 bytes of data:  
Reply from 2404:6800:4002:828::200e: time=49ms  
Reply from 2404:6800:4002:828::200e: time=35ms  
Reply from 2404:6800:4002:828::200e: time=41ms  
  
Ping statistics for 2404:6800:4002:828::200e:  
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 35ms, Maximum = 49ms, Average = 41ms  
Control-C  
^C
```



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(2305CS332)

Date: 25/06/26

```
D:\>ping -w 5000 google.com
```

```
Pinging google.com [2404:6800:4002:828::200e] with 32 bytes of data:
Reply from 2404:6800:4002:828::200e: time=39ms
Reply from 2404:6800:4002:828::200e: time=37ms
Reply from 2404:6800:4002:828::200e: time=36ms
Reply from 2404:6800:4002:828::200e: time=36ms

Ping statistics for 2404:6800:4002:828::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 39ms, Average = 37ms
```

```
D:\>ping -4 google.com
```

```
Pinging google.com [142.250.193.46] with 32 bytes of data:
Reply from 142.250.193.46: bytes=32 time=116ms TTL=110
Reply from 142.250.193.46: bytes=32 time=68ms TTL=110
Reply from 142.250.193.46: bytes=32 time=53ms TTL=110
Reply from 142.250.193.46: bytes=32 time=40ms TTL=110

Ping statistics for 142.250.193.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 116ms, Average = 69ms
```

3. getmac

Description:

----- Details About Command -----

No.	Option	Description
1	getmac	Displays the MAC addresses of all network adapters on the local computer.
2	getmac /v	Displays detailed (verbose) information, including adapter names and transport names.
3	getmac /fo table	Displays the output in Table format. (table, list, or csv are valid formats.)
4	getmac /nh	Suppresses the column headers in the output (useful with table or csv).
5	getmac /fo csv	Displays the output in CSV format.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

```
D:\>getmac

Physical Address      Transport Name
-----
80-2B-F9-2D-D3-C5    \Device\Tcpip_{F4E1251B-9C21-48C2-96F9-DA143A5626CF}
E4-E7-49-68-0D-1D    Media disconnected
00-50-56-C0-00-01    \Device\Tcpip_{6F8E38F2-2D22-4B35-95C9-E0CF87408B32}
00-50-56-C0-00-08    \Device\Tcpip_{91CF5B74-9F49-4515-B273-BE734A73ED25}
```

```
D:\>getmac /v

Connection Name Network Adapter Physical Address      Transport Name
-----
Wi-Fi           Realtek RTL8723 80-2B-F9-2D-D3-C5    \Device\Tcpip_{F4E1251B-9C21-48C2-96F9-DA143A5626CF}
Ethernet        Realtek PCIe Gb E4-E7-49-68-0D-1D    Media disconnected
VMware Network  VMware Virtual  00-50-56-C0-00-01    \Device\Tcpip_{6F8E38F2-2D22-4B35-95C9-E0CF87408B32}
VMware Network  VMware Virtual  00-50-56-C0-00-08    \Device\Tcpip_{91CF5B74-9F49-4515-B273-BE734A73ED25}
```

```
D:\>getmac /fo table

Physical Address      Transport Name
-----
80-2B-F9-2D-D3-C5    \Device\Tcpip_{F4E1251B-9C21-48C2-96F9-DA143A5626CF}
E4-E7-49-68-0D-1D    Media disconnected
00-50-56-C0-00-01    \Device\Tcpip_{6F8E38F2-2D22-4B35-95C9-E0CF87408B32}
00-50-56-C0-00-08    \Device\Tcpip_{91CF5B74-9F49-4515-B273-BE734A73ED25}
```

```
D:\>getmac /nh

80-2B-F9-2D-D3-C5    \Device\Tcpip_{F4E1251B-9C21-48C2-96F9-DA143A5626CF}
E4-E7-49-68-0D-1D    Media disconnected
00-50-56-C0-00-01    \Device\Tcpip_{6F8E38F2-2D22-4B35-95C9-E0CF87408B32}
00-50-56-C0-00-08    \Device\Tcpip_{91CF5B74-9F49-4515-B273-BE734A73ED25}
```

```
D:\>getmac /fo csv
"Physical Address","Transport Name"
"80-2B-F9-2D-D3-C5","\Device\Tcpip_{F4E1251B-9C21-48C2-96F9-DA143A5626CF}"
"E4-E7-49-68-0D-1D","Media disconnected"
"00-50-56-C0-00-01","\Device\Tcpip_{6F8E38F2-2D22-4B35-95C9-E0CF87408B32}"
"00-50-56-C0-00-08","\Device\Tcpip_{91CF5B74-9F49-4515-B273-BE734A73ED25}"
```

4. systeminfo

Description:



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

----- Details About Command -----

No.	Option	Description
1	systeminfo	Displays detailed information about the local computer, including OS, processor, RAM, BIOS, and network details.
2	systeminfo /?	Displays help and all available systeminfo command options.
3	systeminfo /s COMPUTER01	Displays system information for a remote computer.
4	systeminfo /s COMPUTER01 /u DOMAIN\User	Specifies the user account to access the remote computer.
5	systeminfo /s COMPUTER01 /u DOMAIN\User /p Password	Specifies the password for the remote user account.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
D:\>systeminfo

Host Name:                DESKTOP-R2FNR93
OS Name:                   Microsoft Windows 10 Home Single Language
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:          Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:              Multiprocessor Free
Registered Owner:          Hitesh
Registered Organization:
Product ID:                 00327-35114-39536-AAOEM
Original Install Date:      15-08-2024, 13:23:08
System Boot Time:           29-06-2026, 18:23:38
System Manufacturer:       HP
System Model:               HP Laptop 15q-dy0xxx
System Type:                x64-based PC
Processor(s):                1 Processor(s) Installed.
                             [01]: AMD64 Family 21 Model 112 Stepping 0 AuthenticAMD ~1500 Mhz
BIOS Version:               Insyde F.32, 16-07-2020
Windows Directory:          C:\Windows
System Directory:            C:\Windows\system32
Boot Device:                 \Device\HarddiskVolume1
System Locale:                en-us;English (United States)
Input Locale:                00004009
Time Zone:                   (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:       7,646 MB
Available Physical Memory:   4,433 MB
Virtual Memory: Max Size:    8,862 MB
Virtual Memory: Available:   5,157 MB
Virtual Memory: In Use:      3,705 MB
Page File Location(s):       C:\pagefile.sys
Domain:                       WORKGROUP
Logon Server:                 \\DESKTOP-R2FNR93
Hotfix(s):                    28 Hotfix(s) Installed.
                             [01]: KB5066130
                             [02]: KB5042056
                             [03]: KB5011048
                             [04]: KB5015684
                             [05]: KB5020683
                             [06]: KB5033052
                             [07]: KB5072653
                             [08]: KB5071959
                             [09]: KB5014032
```



Date: 25/06/26

```
D:\>systeminfo /?
```

SYSTEMINFO [/S system [/U username [/P [password]]]] [/FO format] [/NH]

Description:
This tool displays operating system configuration information for a local or remote machine, including service pack levels.

Parameter List:

/S	system	Specifies the remote system to connect to.
/U	[domain\]user	Specifies the user context under which the command should execute.
/P	[password]	Specifies the password for the given user context. Prompts for input if omitted.
/FO	format	Specifies the format in which the output is to be displayed. Valid values: "TABLE", "LIST", "CSV".
/NH		Specifies that the "Column Header" should not be displayed in the output. Valid only for "TABLE" and "CSV" formats.
/?		Displays this help message.

Examples:

```
SYSTEMINFO
SYSTEMINFO /?
SYSTEMINFO /S system
SYSTEMINFO /S system /U user
SYSTEMINFO /S system /U domain\user /P password /FO TABLE
SYSTEMINFO /S system /FO LIST
SYSTEMINFO /S system /FO CSV /NH
```

```
D:\>systeminfo /s COUMPTER01
ERROR: The RPC server is unavailable.
```

```
D:\>systeminfo /s COUMPTER01 /u DOMAIN\User
ERROR: The RPC server is unavailable.
```

```
D:\>systeminfo /s COUMPTER01 /u DOMAIN\User /p Password
ERROR: The RPC server is unavailable.
```



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

**5. traceroute/t
racert Description:**

----- Details About Command -----

No.	Option	Description
1	tracert google.com	Traces the route packets take to the destination.
2	tracert -d google.com	Does not resolve IP addresses to hostnames, making the trace faster.
3	tracert -h 20 google.com	Sets the maximum number of hops (here, 20).
4	tracert -w 5000 google.com	Sets the timeout for each reply (here, 5000 ms).
5	tracert -4 google.com	Forces the command to use IPv4.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
D:\>tracert google.com

Tracing route to google.com [2404:6800:4000:100c::8a]
over a maximum of 30 hops:

  1    3 ms    2 ms    2 ms  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
  2    *      *      *      Request timed out.
  3    *      392 ms  24 ms  2405:203:400:100:172:31:3:224
  4   124 ms  102 ms  117 ms  2405:200:801:200::8faa
  5    15 ms   17 ms   16 ms  2405:200:81a:1510:61::7
  6    *      *      *      Request timed out.
  7    21 ms   19 ms   25 ms  2405:200:801:200::8808
  8    24 ms   17 ms   27 ms  2405:200:801:200::8807
  9    24 ms   35 ms   38 ms  2001:4860:1:1::270
```

```
D:\>tracert -d google.com

Tracing route to google.com [2404:6800:4000:100c::8a]
over a maximum of 30 hops:

  1     4 ms     2 ms     2 ms  2409:4090:1001:9403:9632:51ff:fe74:4b5
  2    *        *        *      Request timed out.
  3   315 ms    225 ms    42 ms  2405:203:400:100:172:31:3:224
  4    18 ms     19 ms     17 ms  2405:200:801:200::8faa
  5    15 ms     14 ms     16 ms  2405:200:81a:1510:61::7
  6    *        *        *      Request timed out.
  7    32 ms     23 ms     18 ms  2405:200:801:200::8808
  8    18 ms     15 ms     15 ms  2405:200:801:200::8807
  9    29 ms     23 ms     33 ms  2001:4860:1:1::270
 10   30 ms     24 ms     25 ms  2001:4860:1:1::270
 11 ^C
```



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

```
D:\>tracert -w 5000 google.com

Tracing route to google.com [2404:6800:4000:100c::8a]
over a maximum of 30 hops:

 1      4 ms      3 ms      2 ms  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
 2      *          *          *      Request timed out.
 3     79 ms     39 ms     38 ms  2405:203:400:100:172:31:3:224
 4    128 ms     69 ms     21 ms  2405:200:801:200::8faa
 5   1252 ms     36 ms     38 ms  2405:200:81a:1510:61::7
^C

D:\>tracert -h 20 google.com

Tracing route to google.com [2404:6800:4000:100c::8a]
over a maximum of 20 hops:

 1      3 ms      2 ms      2 ms  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
 2      *          *          *      Request timed out.
 3    101 ms     28 ms     40 ms  2405:203:400:100:172:31:3:224
 4     76 ms     55 ms     47 ms  2405:200:801:200::8faa
 5     64 ms     29 ms     94 ms  2405:200:81a:1510:61::7
 6      *          *          *      Request timed out.
 7    392 ms     272 ms    362 ms  2405:200:801:200::8808
 8     74 ms     105 ms    120 ms  2405:200:801:200::8807
 9    313 ms     195 ms     31 ms  2001:4860:1:1::270

D:\>tracert -4 google.com

Tracing route to google.com [192.178.173.139]
over a maximum of 30 hops:

 1      4 ms      2 ms      2 ms  jiofiber.local.html [192.168.31.1]
 2      7 ms      3 ms      3 ms  192.0.0.1
 3      *          *          *      Request timed out.
 4     69 ms     393 ms     688 ms  192.0.0.1
 5    231 ms      46 ms      27 ms  192.0.0.1
 6     20 ms      17 ms      18 ms  192.0.0.1
```

6. netstat

Description:

----- Details About Command -----

No.	Option	Description
1	netstat -a	Displays all active connections and listening ports.
2	netstat -n	Displays addresses and port numbers in numerical form.
3	netstat -o	Displays the Process ID (PID) associated with each connection.
4	netstat -r	Displays the IP routing table.
5	netstat -e	Displays Ethernet statistics, such as bytes sent and received.

Implementation:



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
D:\>netstat -a

Active Connections

Proto Local Address          Foreign Address         State
TCP   0.0.0.0:135             DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:445             DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:902             DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:912             DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:5040            DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:7680            DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49664           DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49665           DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49666           DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49667           DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49668           DESKTOP-R2FNR93:0      LISTENING
TCP   0.0.0.0:49669           DESKTOP-R2FNR93:0      LISTENING
TCP   169.254.5.118:139       DESKTOP-R2FNR93:0      LISTENING
TCP   169.254.246.19:139      DESKTOP-R2FNR93:0      LISTENING
TCP   192.168.31.120:139      DESKTOP-R2FNR93:0      LISTENING
TCP   192.168.31.120:50711    20.50.73.14:https      TIME_WAIT
^C
```

```
D:\>netstat -n

Active Connections

Proto Local Address          Foreign Address         State
TCP   192.168.31.120:50711    20.50.73.14:443         TIME_WAIT
TCP   192.168.31.120:50713    20.190.146.37:443       TIME_WAIT
TCP   192.168.31.120:50715    104.208.16.92:443       TIME_WAIT
TCP   192.168.31.120:50716    20.44.10.122:443        ESTABLISHED
TCP   192.168.31.120:50717    98.70.192.224:443       ESTABLISHED
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50654 [2603:1040:a06:6::2]:443 ESTABLISHED
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50684 [2405:200:1609:1731::312c:70d4]:443 CLOSE_WAIT
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50686 [2603:1046:c04:83b::2]:443 ESTABLISHED
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50688 [2603:1061:14:150::1]:443 CLOSE_WAIT
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50701 [2600:140f:d800:180::ae0]:443 CLOSE_WAIT
TCP   [2409:4090:1001:9403:2192:8a40:20f:9012]:50714 [2600:14e1:1c:6d::]:80  TIME_WAIT
```

```
D:\>netstat -o

Active Connections

Proto Local Address          Foreign Address         State      PID
TCP   192.168.31.120:50711    20.50.73.14:https      TIME_WAIT  0
TCP   192.168.31.120:50713    20.190.146.37:https     TIME_WAIT  0
^C
```



Date: 25/06/26

```
D:\>netstat -r
=====
Interface List
 5...e4 e7 49 68 0d 1d .....Realtek PCIe GbE Family Controller
14...82 2b f9 2d d3 c5 .....Microsoft Wi-Fi Direct Virtual Adapter #3
 6...80 2b f9 2d d3 c5 .....Microsoft Wi-Fi Direct Virtual Adapter #4
15...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
17...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
22...80 2b f9 2d d3 c5 .....Realtek RTL8723DE 802.11b/g/n PCIe Adapter
 1.....Software Loopback Interface 1
=====

D:\>netstat -e
Interface Statistics

              Received              Sent
Bytes          7571938          4615534
Unicast packets    11860          11210
Non-unicast packets 14364          11600
Discards           0              0
Errors             0              0
Unknown protocols  0
```

7. nslookup

Description:

----- Details About Command -----

No.	Option	Description
1	nslookup google.com	Displays the IP address of the specified domain.
2	nslookup -type=A google.com	Displays the IPv4 (A) record of the domain.
3	nslookup -type=MX google.com	Displays the Mail Exchange (MX) records of the domain.
4	nslookup -type=NS google.com	Displays the Name Server (NS) records of the domain.
5	nslookup -type=AAAA google.com	Displays the IPv6 (AAAA) record of the domain.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



Date: 25/06/26

```
D:\>nslookup google.com
Server: jiofiber.local.html
Address: 192.168.31.1

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4013:804::8a
           2404:6800:4013:804::66
           2404:6800:4013:804::65
           2404:6800:4013:804::8b
           142.250.70.110

D:\>nslookup -type=A google.com
Server: jiofiber.local.html
Address: 192.168.31.1

Non-authoritative answer:
Name: google.com
Address: 142.250.70.110

D:\>nslookup -type=MX google.com
Server: jiofiber.local.html
Address: 192.168.31.1

Non-authoritative answer:
google.com MX preference = 10, mail exchanger = smtp.google.com
```



Date: 25/06/26

```
D:\>nslookup -type=NS google.com
Server: jiofiber.local.html
Address: 192.168.31.1

Non-authoritative answer:
google.com      nameserver = ns4.google.com
google.com      nameserver = ns1.google.com
google.com      nameserver = ns3.google.com
google.com      nameserver = ns2.google.com

ns2.google.com  internet address = 216.239.34.10
ns2.google.com  AAAA IPv6 address = 2001:4860:4802:34::a
ns4.google.com  internet address = 216.239.38.10
ns4.google.com  AAAA IPv6 address = 2001:4860:4802:38::a
ns1.google.com  internet address = 216.239.32.10
ns1.google.com  AAAA IPv6 address = 2001:4860:4802:32::a
ns3.google.com  internet address = 216.239.36.10
ns3.google.com  AAAA IPv6 address = 2001:4860:4802:36::a

D:\>nslookup -type=NS google.com
Server: jiofiber.local.html
Address: 192.168.31.1

Non-authoritative answer:
google.com      nameserver = ns4.google.com
google.com      nameserver = ns2.google.com
google.com      nameserver = ns3.google.com
google.com      nameserver = ns1.google.com

ns4.google.com  internet address = 216.239.38.10
ns4.google.com  AAAA IPv6 address = 2001:4860:4802:38::a
ns2.google.com  internet address = 216.239.34.10
ns2.google.com  AAAA IPv6 address = 2001:4860:4802:34::a
ns3.google.com  internet address = 216.239.36.10
ns3.google.com  AAAA IPv6 address = 2001:4860:4802:36::a
ns1.google.com  internet address = 216.239.32.10
ns1.google.com  AAAA IPv6 address = 2001:4860:4802:32::a
```



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(2305CS332)

Date: 25/06/26

8. hostname

Description:

----- Details About Command -----

No.	Option	Description
1	hostname	Displays the computer name.
2	hostname > hostname.txt	Saves the computer name to a text file.
3	hostname >> hostname.txt	Appends the computer name to a text file.
4	hostname clip	Copies the computer name to the clipboard.
5	hostname /?	Attempts to display help (on most Windows versions it simply prints the hostname because no command-line options are supported).

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
D:\>hostname
DESKTOP-R2FNR93

D:\>hostname > hostname.txt

D:\>hostname >> hostname.txt

D:\>hostname | clip

D:\>hostname /?

Prints the name of the current host.
hostname
```



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(2305CS332)

Date: 25/06/26

9. pathping

Description:

----- Details About Command -----

No.	Option	Description
1	pathping google.com	Traces the route to the destination and measures packet loss at each hop.
2	pathping -n google.com	Does not resolve IP addresses to hostnames, making the command faster.
3	pathping -h 20 google.com	Sets the maximum number of hops (here, 20).
4	pathping -w 5000 google.com	Sets the timeout for each reply to 5000 milliseconds.
5	pathping -q 50 google.com	Sends 50 query packets to each hop for more accurate statistics.

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----



Date: 25/06/26

```
D:\>pathping google.com

Tracing route to google.com [2404:6800:4013:804::8b]
over a maximum of 30 hops:
  0  DESKTOP-R2FNR93.lan [2409:4090:1001:9403:2192:8a40:20f:9012]
  1  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
  2  ^C
D:\>pathping -n google.com

Tracing route to google.com [2404:6800:4013:804::8b]
over a maximum of 30 hops:
  0  2409:4090:1001:9403:2192:8a40:20f:9012
  1  2409:4090:1001:9403:9632:51ff:fe74:4b5
  2  ^C
D:\>pathping -h 20 google.com

Tracing route to google.com [2404:6800:4013:804::8b]
over a maximum of 20 hops:
  0  DESKTOP-R2FNR93.lan [2409:4090:1001:9403:2192:8a40:20f:9012]
  1  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
  2  * * *
Computing statistics for 25 seconds...
^C
D:\>pathping -w 5000 google.com

Tracing route to google.com [2404:6800:4013:804::8b]
over a maximum of 30 hops:
  0  DESKTOP-R2FNR93.lan [2409:4090:1001:9403:2192:8a40:20f:9012]
  1  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
  2  ^C
D:\>pathping -q 50 google.com

Tracing route to google.com [2404:6800:4013:804::8b]
over a maximum of 30 hops:
  0  DESKTOP-R2FNR93.lan [2409:4090:1001:9403:2192:8a40:20f:9012]
  1  jiofiber.local.html [2409:4090:1001:9403:9632:51ff:fe74:4b5]
  2  * * *
Computing statistics for 12 seconds...
^C
```

10. arp

Description:

----- Details About Command -----

No.	Option	Description
1	arp -a	Displays the current ARP cache table for all interfaces.
2	arp -d *	Deletes all entries from the ARP cache.
3	arp -d *	Deletes the ARP entry for the specified IP address.
4	arp -s 192.168.1.100 00-11-22-33-44-55	Adds a static ARP entry with the specified IP and MAC address.
5	arp -g	Displays the ARP cache table (same as arp -a).



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MCA Semester 3th | Practical Assignment | Computer Networks
(2305CS332)

Date: 25/06/26

Implementation:

----- Screenshot Only (Execute command on command prompt / terminal) -----

```
D:\>arp -a

Interface: 169.254.246.19 --- 0xf
Internet Address      Physical Address      Type
169.254.255.255       ff-ff-ff-ff-ff-ff     static
192.168.111.254        00-50-56-fa-fc-19     dynamic
224.0.0.22             01-00-5e-00-00-16     static
224.0.0.251            01-00-5e-00-00-fb     static
224.0.0.252            01-00-5e-00-00-fc     static
239.255.255.250        01-00-5e-7f-ff-fa     static
255.255.255.255        ff-ff-ff-ff-ff-ff     static

D:\>arp -d *
The ARP entry deletion failed: The requested operation requires elevation.

D:\>arp -d *
The ARP entry deletion failed: The requested operation requires elevation.

D:\>arp -s 192.168.1.100 00-11-22-33-44-55
The ARP entry addition failed: The requested operation requires elevation.

D:\>arp -g

Interface: 169.254.246.19 --- 0xf
Internet Address      Physical Address      Type
169.254.255.255       ff-ff-ff-ff-ff-ff     static
192.168.111.254        00-50-56-fa-fc-19     dynamic
224.0.0.22             01-00-5e-00-00-16     static
224.0.0.251            01-00-5e-00-00-fb     static
224.0.0.252            01-00-5e-00-00-fc     static
239.255.255.250        01-00-5e-7f-ff-fa     static
255.255.255.255        ff-ff-ff-ff-ff-ff     static
```

**Lab Practical #02:**

Study of different network devices in detail.

Practical Assignment #02:

1. Give difference between below network devices.
 - Hub and Switch
 - Switch and Router
2. Working of below network devices:
 - Repeater
 - Modem(DSL and ADSL)
 - Hub
 - Bridge
 - Switch
 - Router
 - Gateway

Hub and Switch

No.	Hub	Switch
1	Hub sends data to all connected devices.	Switch sends data only to the intended device.
2	Works on the Physical Layer (Layer 1).	Works on the Data Link Layer (Layer 2).
3	Does not store MAC addresses.	Stores MAC addresses in a MAC table.
4	Slower and creates more network traffic.	Faster and reduces network traffic.
5	Less secure because data is broadcast to all devices.	More secure because data is sent only to the destination device.

Switch and Router

No.	Switch	Router
1	Connects devices within the same network (LAN).	Connects different networks (LAN to WAN/Internet).
2	Works on the Data Link Layer (Layer 2).	Works on the Network Layer (Layer 3).
3	Uses MAC addresses to forward data.	Uses IP addresses to route data.
4	Used to create and manage a local network.	Used to provide Internet access and connect multiple networks.
5	Faster for communication inside a LAN.	Slower than a switch because it performs routing decisions.

Router and Gateway

No.	Router	Gateway



1	Connects different networks.	Connects networks that use different protocols.
2	Uses IP addresses to route data.	Translates data between different network protocols.
3	Works on the Network Layer (Layer 3).	Can work on all OSI layers, depending on its function.
4	Mainly used to forward data packets between networks.	Used to enable communication between dissimilar networks or applications.
5	Example: Wi-Fi router connecting a home network to the Internet.	Example: Email gateway or VoIP gateway connecting different systems.

Working of below network devices:

1. Repeater

- ✓ A repeater is a network device that receives weak or damaged signals, regenerates (strengthens) them, and retransmits them.
- ✓ It is used to extend the communication distance between network devices.
- ✓ It works on the **Physical Layer (Layer 1)** of the OSI model.
- ✓ It does not filter or modify data; it only boosts the signal strength.

2. Modem

- ✓ A modem converts **digital signals** from a computer into **analog signals** for transmission over telephone lines and converts them back to digital at the receiving end.
- ✓ **DSL (Digital Subscriber Line)** provides internet access over telephone lines and supports both voice calls and internet simultaneously. It is commonly used in **small businesses and offices** where equal upload and download speeds may be needed.
- ✓ **ADSL (Asymmetric Digital Subscriber Line)** is a type of DSL in which the **download speed is higher than the upload speed**. It is mainly used in **homes and residential areas** because users usually download more data than they upload.
- ✓ A modem establishes the connection between the user's device and the **Internet Service Provider (ISP)**.

3. Hub

- ✓ A hub is a network device that connects multiple computers or devices in a **Local Area Network (LAN)**.
- ✓ It receives data from one device and **broadcasts it to all connected devices**, regardless of the destination.
- ✓ It works on the **Physical Layer (Layer 1)** of the OSI model.
- ✓ It does not store MAC addresses or make forwarding decisions, making it slower and less secure than a switch.



- ✓ It is mainly used in **small or temporary networks** and is rarely used today because switches provide better performance.

4. Bridge

- ✓ A bridge is a network device that **connects two or more LAN segments** and allows them to communicate.
- ✓ It works on the **Data Link Layer (Layer 2)** of the OSI model.
- ✓ It uses **MAC addresses** to identify the destination device and forwards data only to the required network segment.
- ✓ It reduces unnecessary network traffic by filtering data packets.
- ✓ It is commonly used in **small to medium-sized LANs** to improve network performance and reduce collisions.

5. Switch

- ✓ A switch is a network device that connects multiple devices in a **Local Area Network (LAN)**.
- ✓ It works on the **Data Link Layer (Layer 2)** of the OSI model.
- ✓ It uses **MAC addresses** to identify the destination device and sends data **only to the intended device**.
- ✓ It reduces network traffic and collisions, providing faster and more efficient communication than a hub.
- ✓ It is commonly used in **homes, offices, schools, and business networks** to connect computers, printers, and other network devices.

6. Router

- ✓ A router is a network device that **connects two or more different networks**, such as a LAN and the Internet.
- ✓ It works on the **Network Layer (Layer 3)** of the OSI model.
- ✓ It uses **IP addresses** to determine the best path and forward data packets to their destination.
- ✓ It allows multiple devices to **share a single Internet connection** and can provide features like **NAT, DHCP, and basic firewall protection**.
- ✓ It is commonly used in **homes, schools, offices, and businesses** to connect local networks to the Internet.

7. Gateway

- ✓ A gateway is a network device that **connects networks using different communication protocols**.
- ✓ It acts as a **translator**, converting data from one protocol or format to another so different networks can communicate.



- ✓ It can operate on **multiple OSI layers**, depending on its function.
- ✓ It enables communication between **dissimilar networks**, such as a private LAN and the Internet or between different software systems.
- ✓ It is commonly used in **large organizations, cloud services, IoT networks, and enterprise networks** where different types of networks or protocols need to work together.

Lab Practical #03:

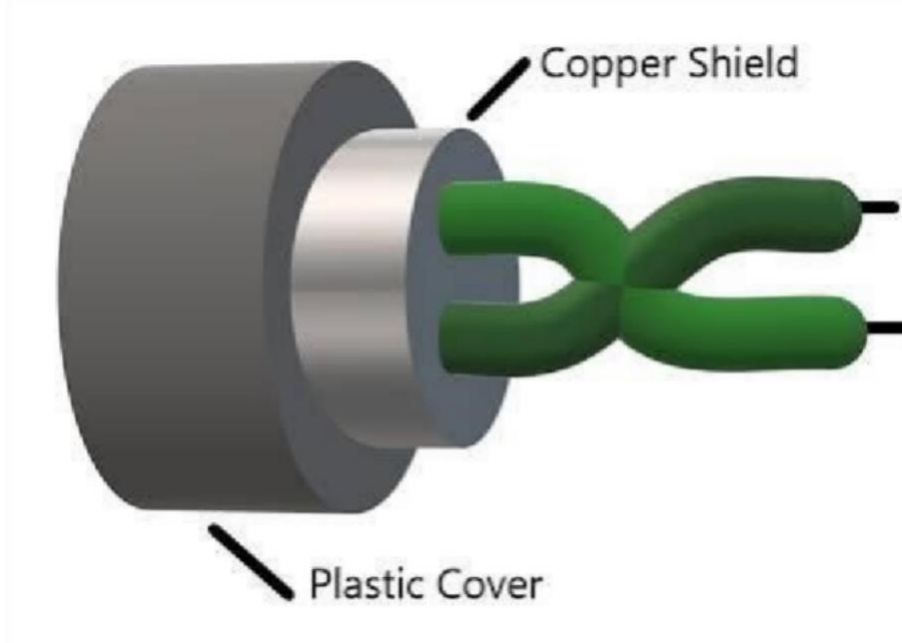
Study of different types of network cables & connectors and crimping a LAN.

Practical Assignment #03:

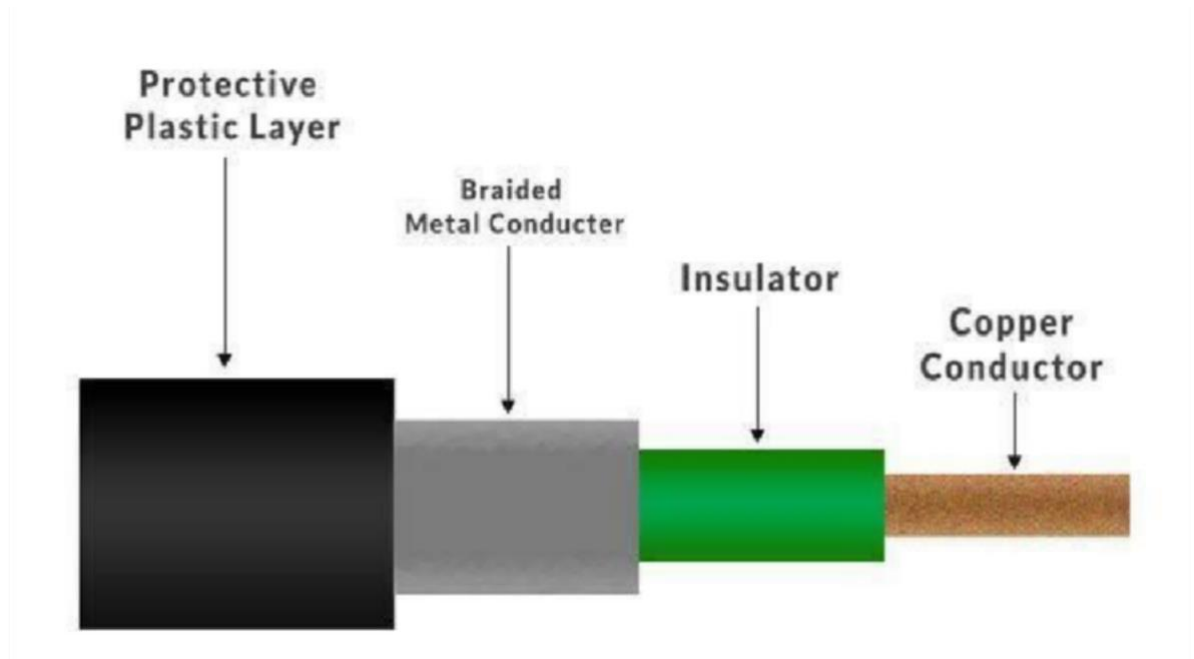
1. List various networks cable. Also, write short description.
2. Difference between guided and unguided media.
3. Give cross-wired cable and straight through cable diagram (Color Code wise).

1. List various networks cable and connectors. Also, write short description.

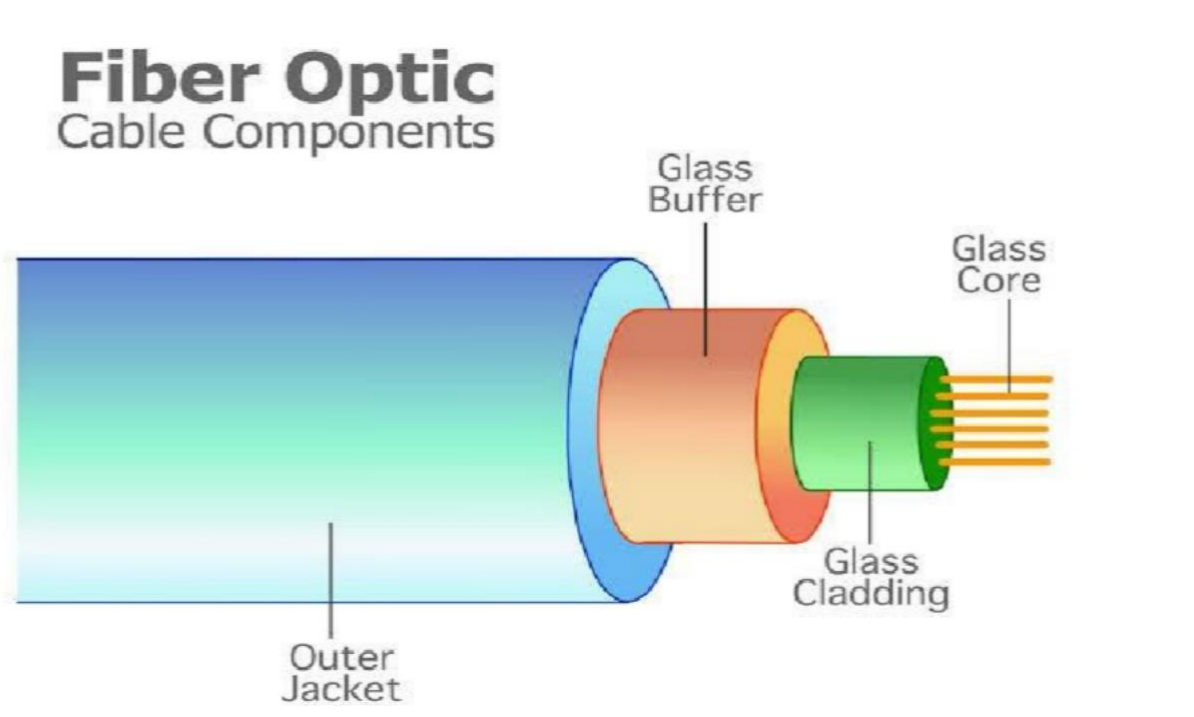
- **Twisted Pair Cable (UTP/STP):**
- **Description:** The most common cable used in LANs. It consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference (EMI). It comes in two types: **Unshielded Twisted Pair (UTP)** (common in homes/offices) and **Shielded Twisted Pair (STP)** (has protective foil for high-interference environments). Commonly categorized as Cat5e, Cat6, or Cat6a.
- **Diagram:**



- **Coaxial Cable:**
- **Description:** Contains a single copper conductor at its core, surrounded by a plastic insulator, a braided metal shield to block noise, and an outer plastic jacket. Historically used for early computer networks (Thicknet/Thinnet) but today primarily used for cable television (TV) and broadband internet modems.
- **Diagram:**



- **Fiber Optic Cable:**
- **Description:** Transmits data as pulses of light through tiny strands of glass or plastic (core). Because it uses light instead of electricity, it is completely immune to electromagnetic interference, offers incredibly high bandwidth, and can carry data over massive distances. It is split into **Single-mode** (long distance) and **Multi-mode** (shorter distances like data centers).
- **Diagram:**



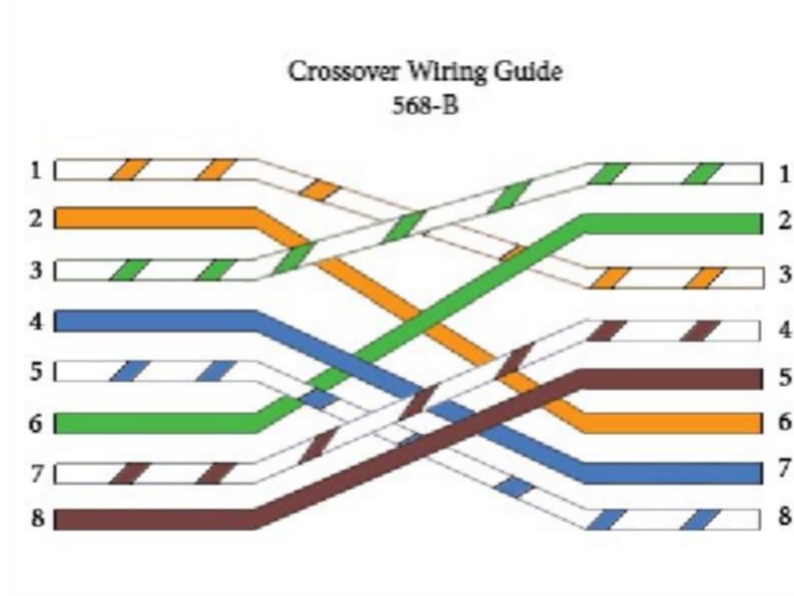
2. Difference between guided and unguided media.



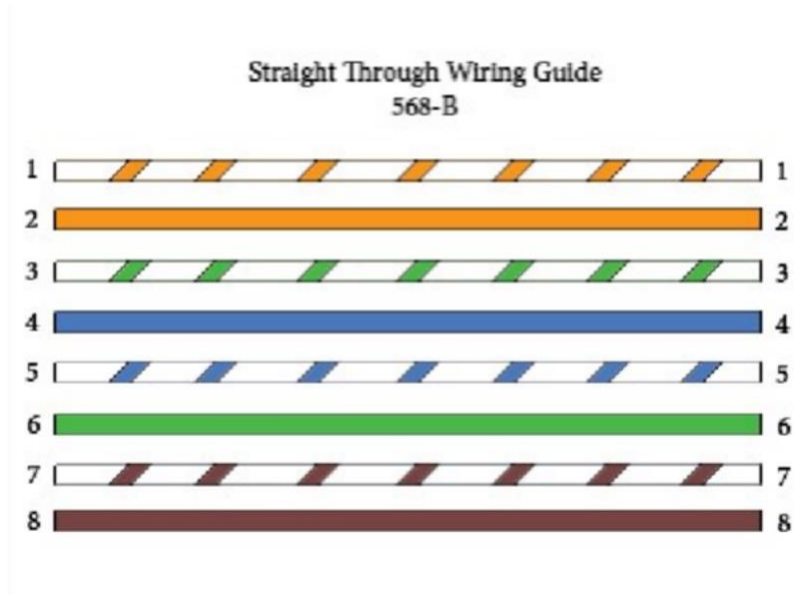
Feature	Guided Media (Wired)	Unguided Media (Wireless)
Definition	Physical wired pathways through which signals are directed and contained.	Wireless systems that transmit electromagnetic waves without physical drivers.
Medium Type	Solid/Physical (Copper, Glass).	Free Space / Atmosphere (Air, Vacuum).
Signal Direction	Signals are guided along a specific physical path.	Signals are broadcasted in all directions through the air.
Examples	Twisted Pair (Cat6), Coaxial, Fiber Optic cables.	Wi-Fi, Bluetooth, Satellite, Infrared, Radio waves.
Interference	Generally lower (can be shielded against EMI).	Higher (prone to weather, obstacles, and cross-channel noise).
Cost S Installation	Installation is laborintensive; cost scales with distance.	Initial setup can be high, but scales easily without adding physical wires.

3. Give cross-wired cable and straight through cable diagram (Color Code wise).

a) Cross-wired Cable Diagram (Color Code)



b) Straight Through Cable Diagram (Color Code)



Lab Practical #04:

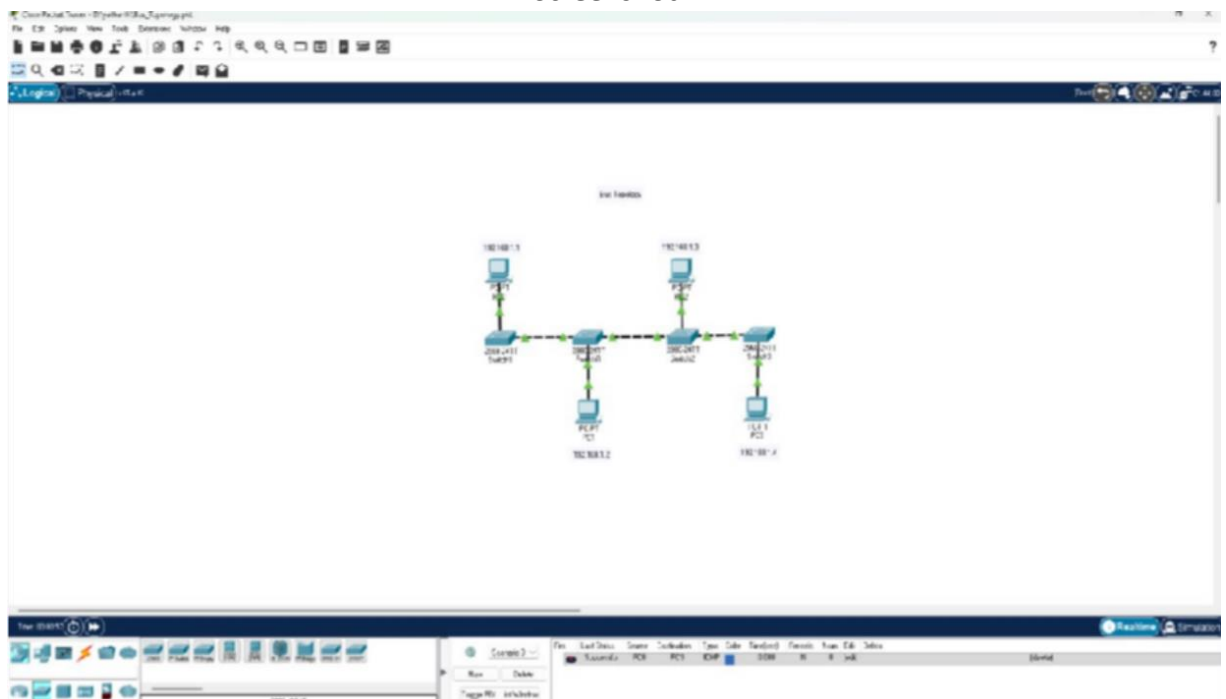
Installation of Network Simulator (Packet Tracer) and Implement different LAN topologies. **Practical Assignment #04:**

1. Create a simple network with switch and two or more pc. Also check connectivity between them using ping command or PDU utility.
2. Implement different topologies in packet tracer.
 - a. Bus
 - b. Ring
 - c. Star
 - d. Mesh
 - e. Tree

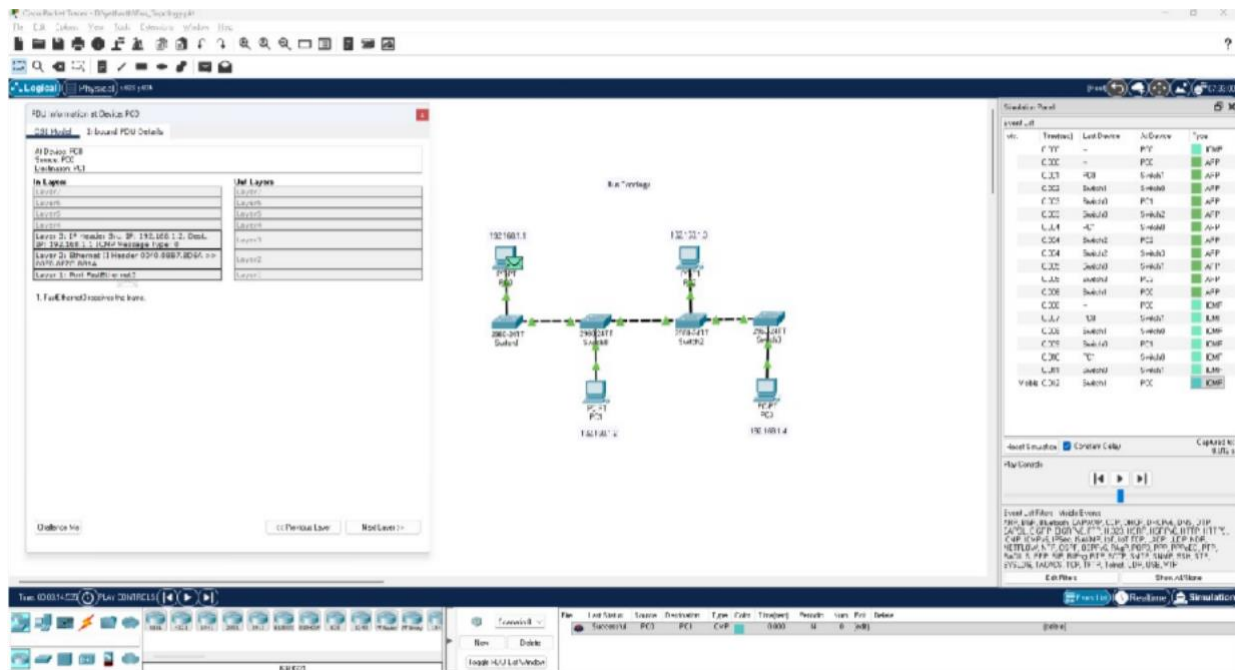
Instructions:

1. Topology-wise Screenshot.
2. Mention IP address of each pc as label.
3. Ping command or PDU screenshot between two pcs.

Answers:

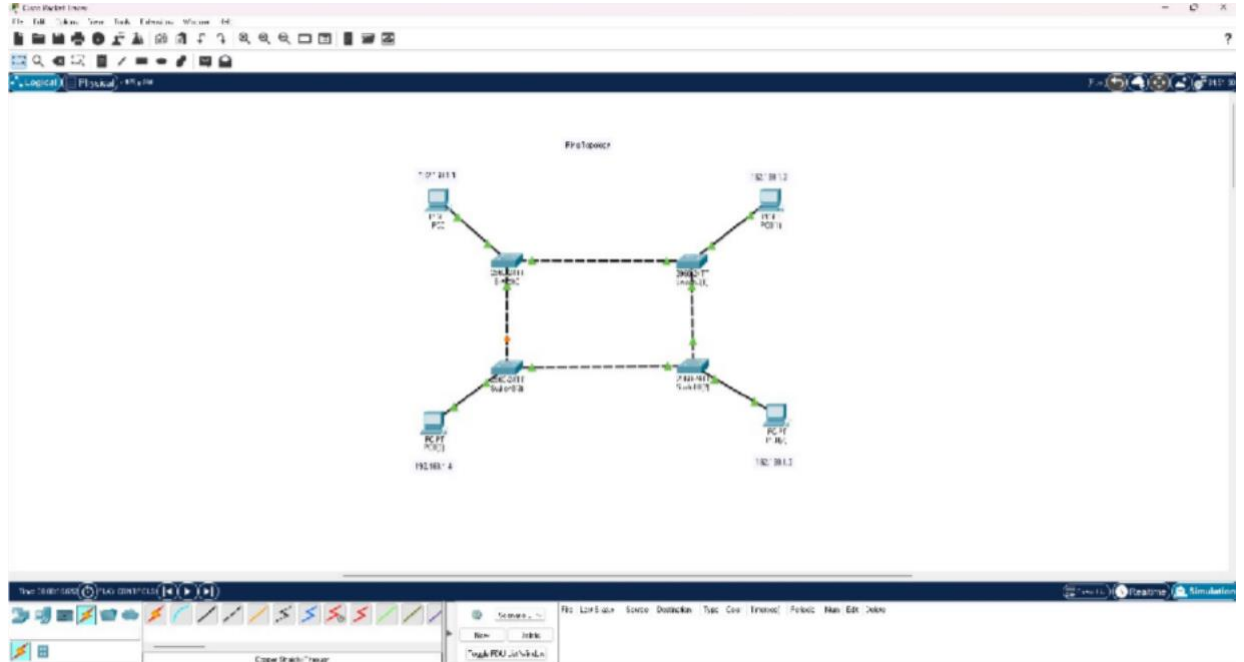
1. Bus topology**Screenshot-1**

Screenshot-2

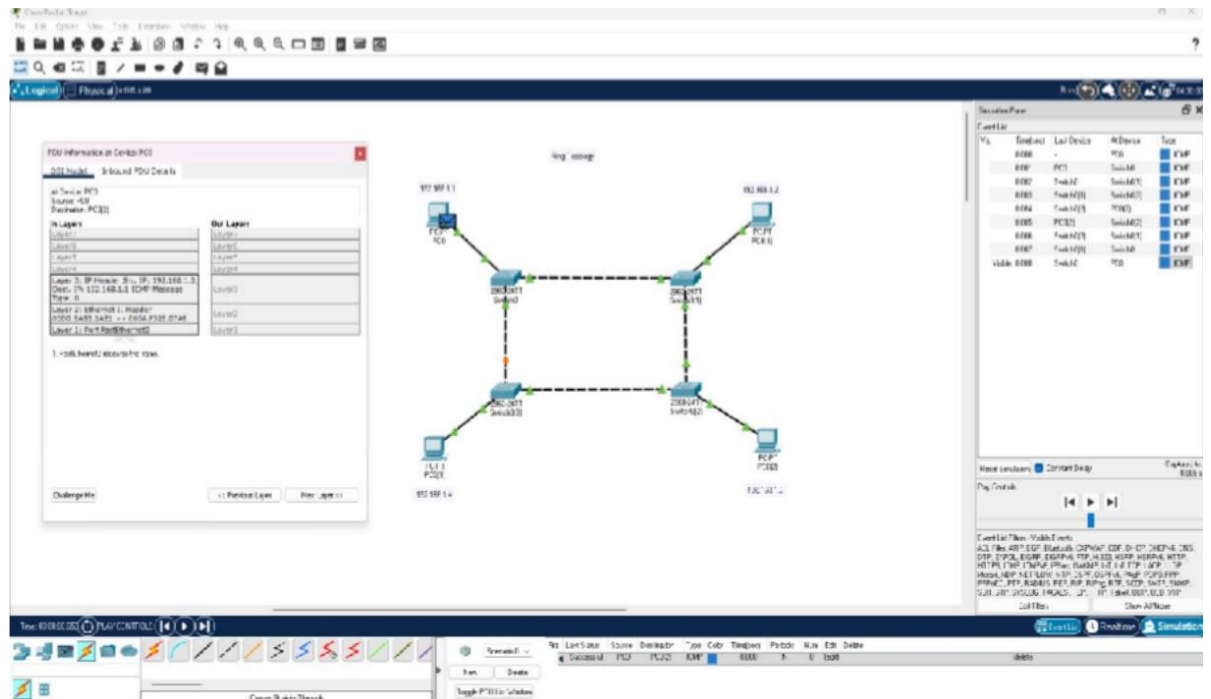


2. Ring topology

Screenshot-1

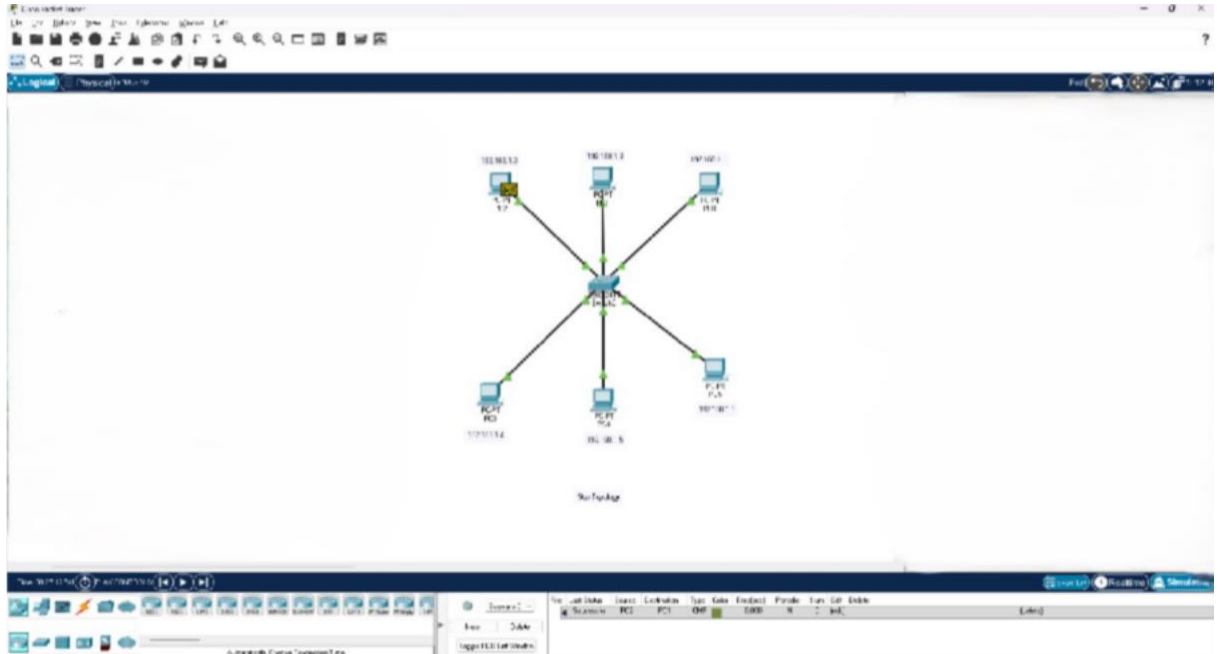


Screenshot-2

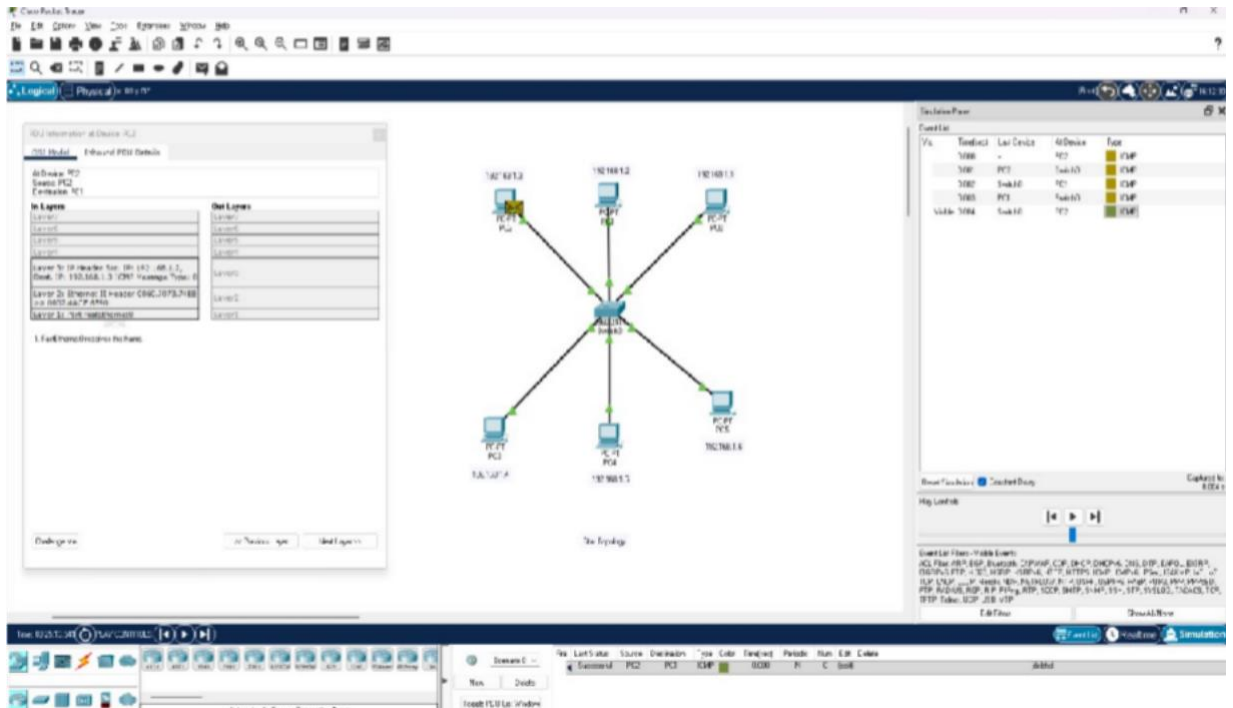


3. Star topology

Screenshot-1

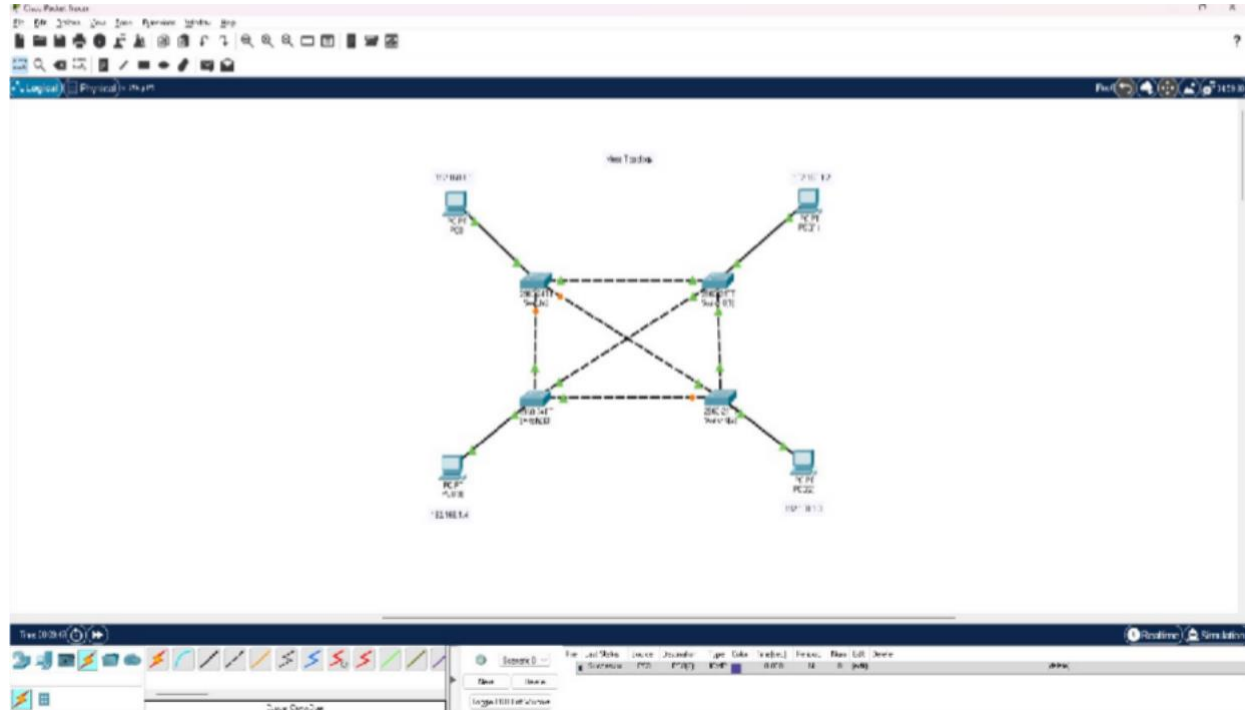


Screenshot-2

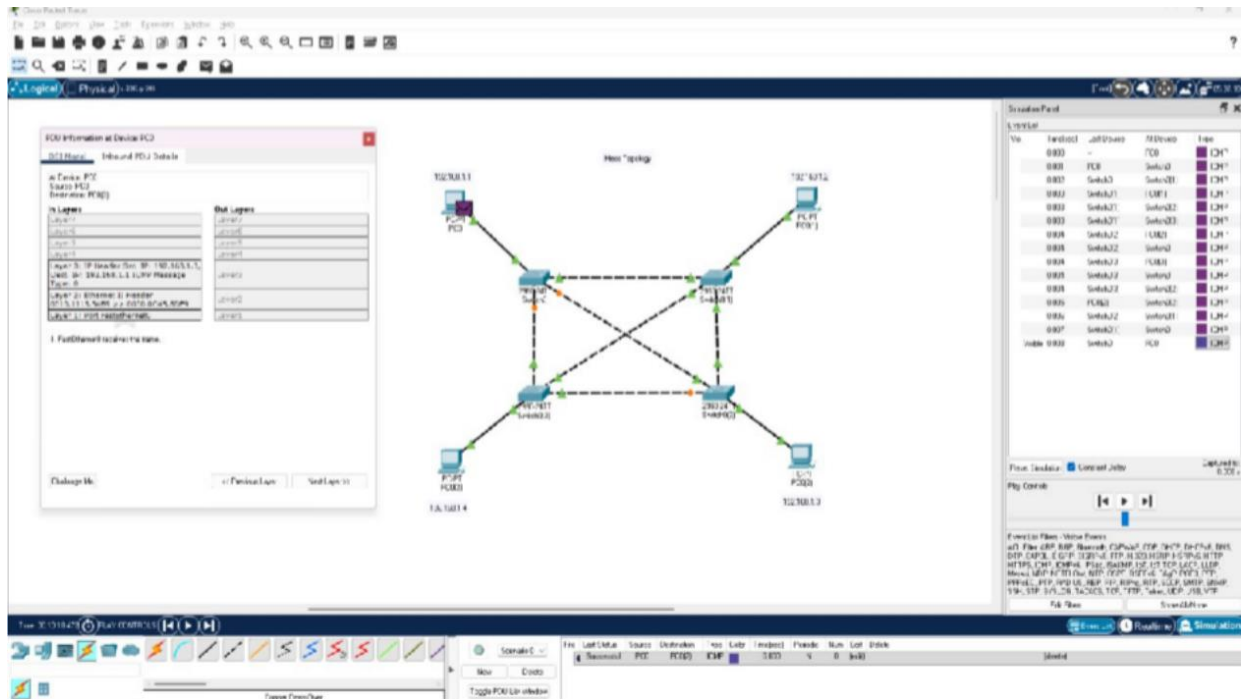


4. Mesh topology

Screenshot-1

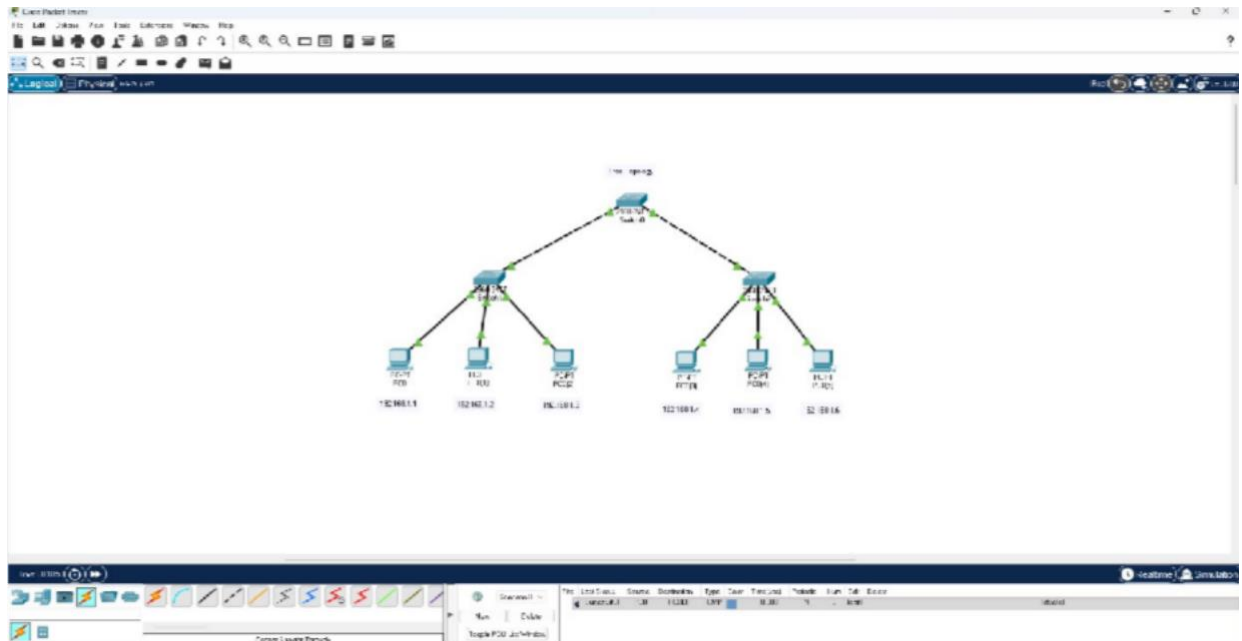


Screenshot-2

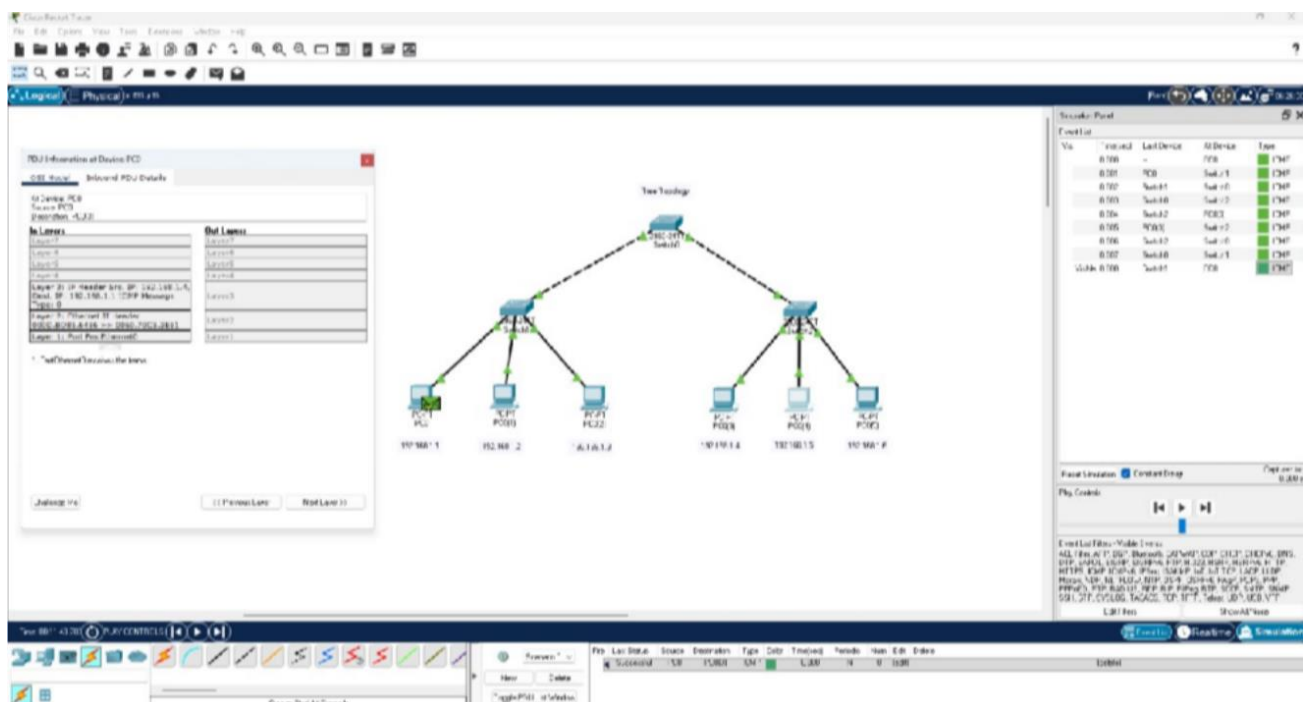


5. Tree topology

Screenshot-1



Screenshot-2





Lab Practical #05:

Study the concept of VLAN using packet tracer. **Practical**

Assignment #05:

1. Implement the different network structures in VLAN and VLAN trunking. Also check connectivity between them using ping command or PDU utility.

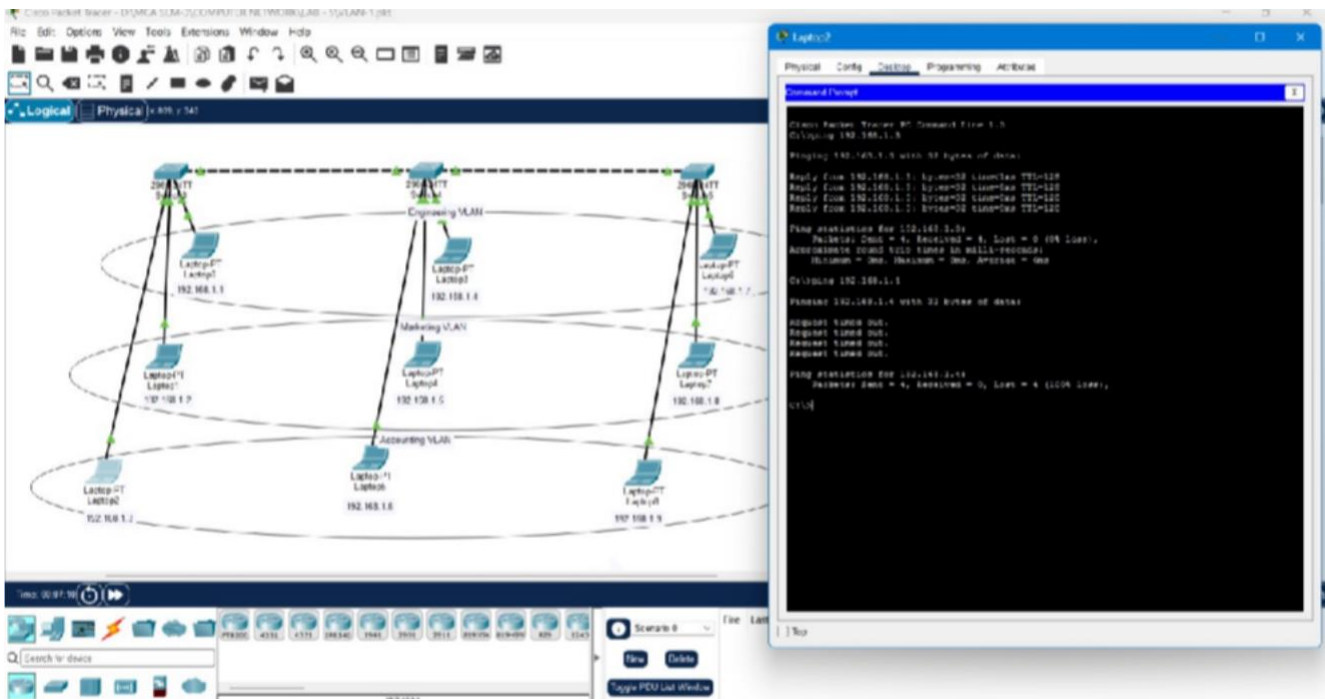
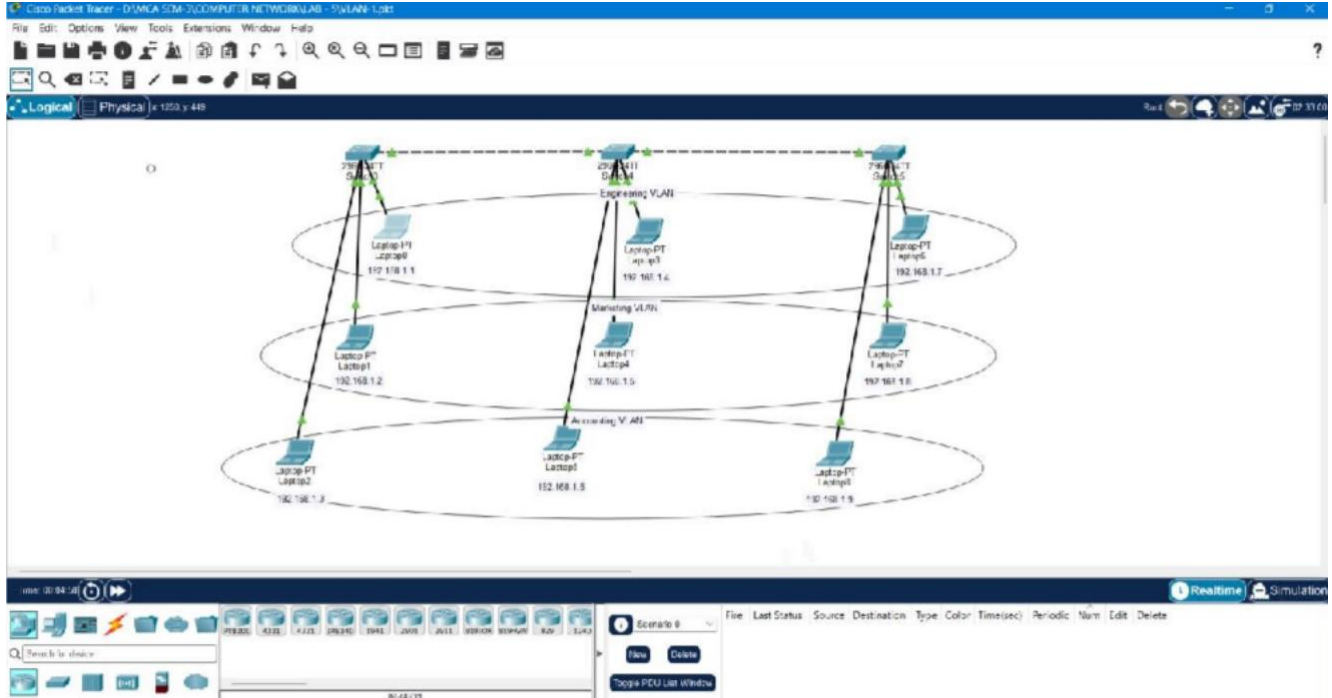
Instructions:

1. Different VLANs configuration setup screenshot. (VLAN example given by lab faculty)
2. Write steps to create VLANs in packet tracer.
 - Topology: Connect PC0 (Port 1) and PC1 (Port 2) to a Switch.
 - Create VLANs: Open Switch > Config tab > VLAN Database. Add VLAN 10 (Sales) and VLAN 20 (IT).
 - Assign Ports: In Switch Config, select FastEthernet0/1 and set VLAN to 10. Select FastEthernet0/2 and set VLAN to 20.
 - Set IP Addresses:
 - PC0: IP 192.168.1.10 / Mask 255.255.255.0
 - PC1: IP 192.168.1.20 / Mask 255.255.255.0
 - Test: Open PC0 Command Prompt and ping 192.168.1.20. (*Fails because they are on different VLANs*).
3. Mention IP address of each pc as label.
4. Ping command or PDU screenshot between two VLANs.



Date: 6/08/2026

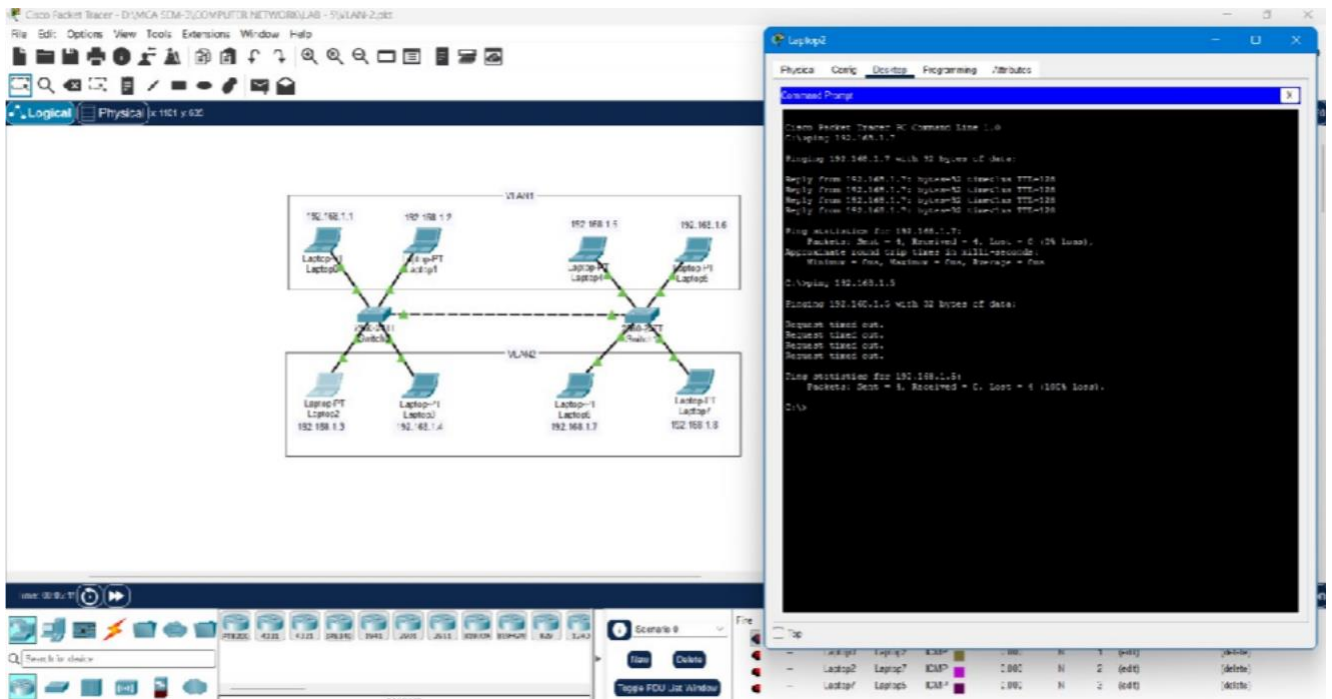
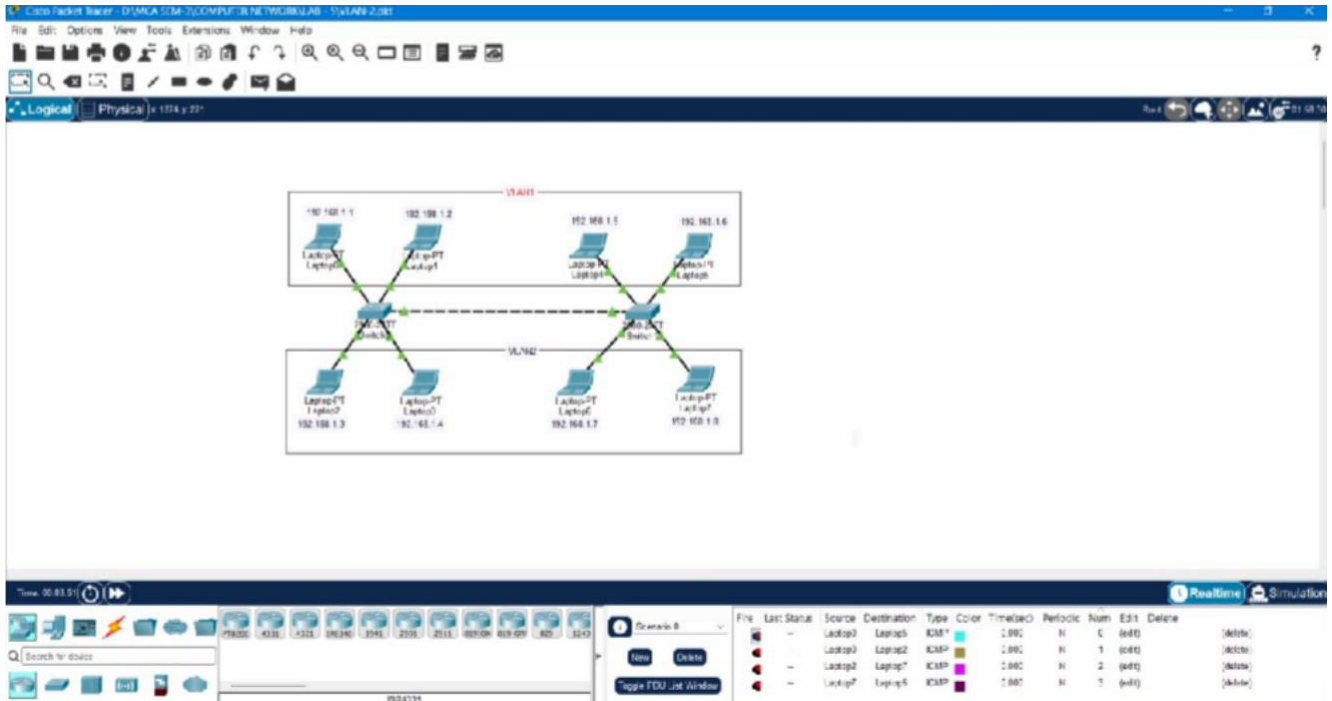
Screenshot1





Date: 6/08/2026

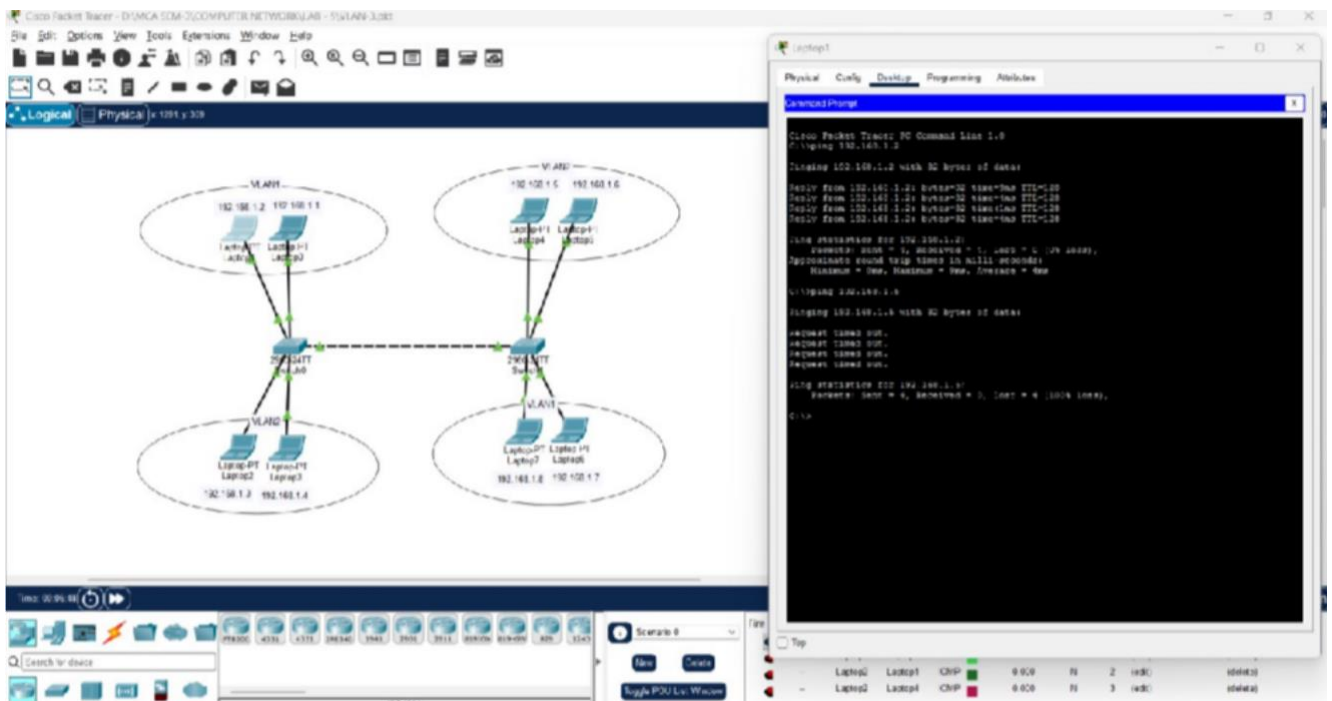
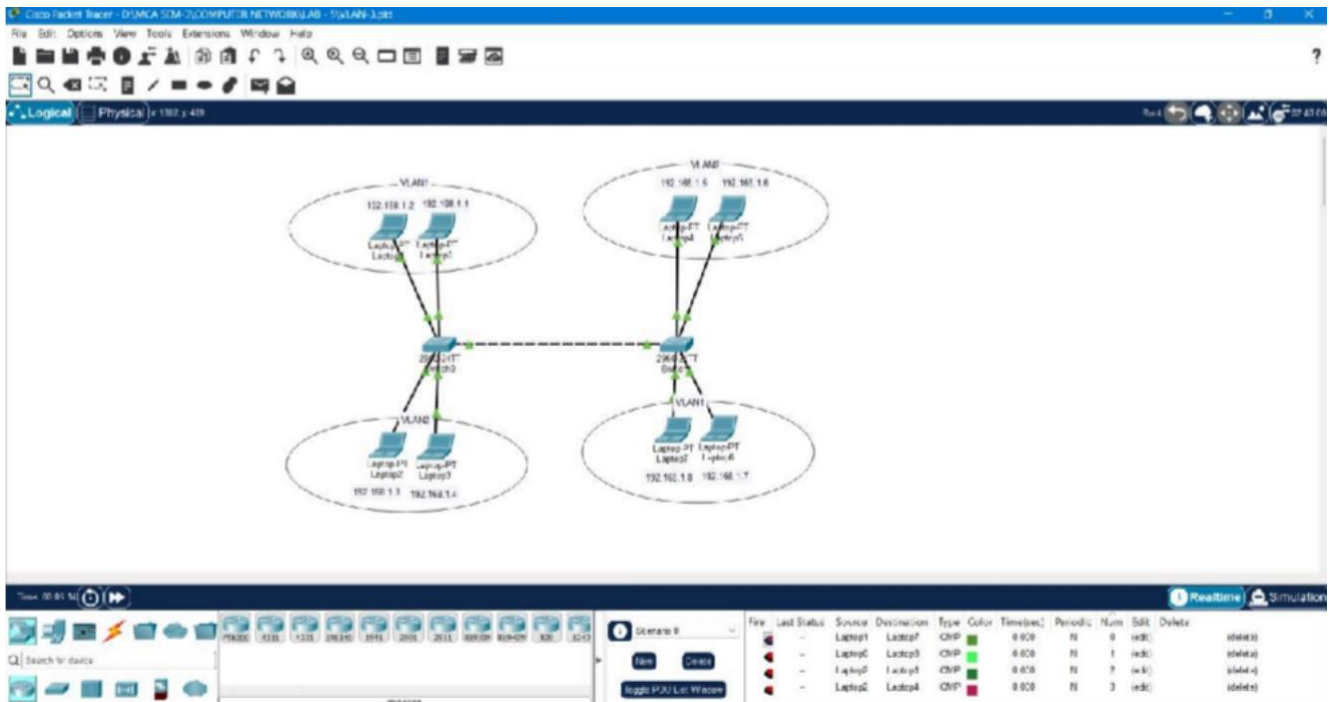
Screenshot2:



Screenshot3:



Date: 6/08/2026





Date: 27/08/2026

Lab Practical #06:

Study the application layer protocol DNS, DHCP.

1. Implement the application layer protocol DNS and DHCP. Also check connectivity between them using ping command or PDU utility.

Instructions:

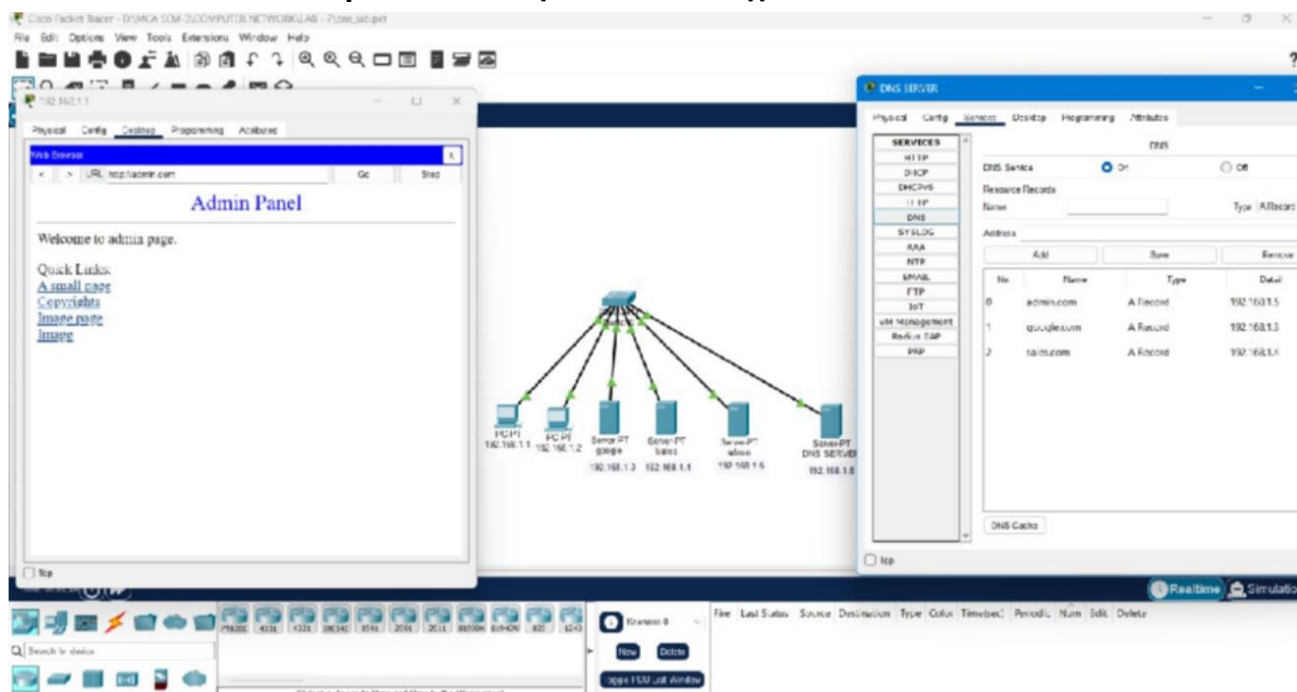
1. Protocol-wise configuration setup screenshot.
2. Mention IP address of each pc as label.

Implementation:

DNS :

create a network using 2 pc 1 switch and 4 server where 3 server act as web server for following

- 1.) sales.com 2.)google.edu 3.)admin.org and the 4th acts as a DNS Server then open all three websites through web browser of pc. (Attach screenshot 1. Showing DNS database 2. Website open from PC(All 3 website))





Date: 27/08/2026

DHCP:

Implement 3 Topology and configure ip of pc in the topology through the DHCP service of the server

1.) Mesh Topology using DHCP

2.) Star Topology using DHCP

3.) Tree Topology using DHCP

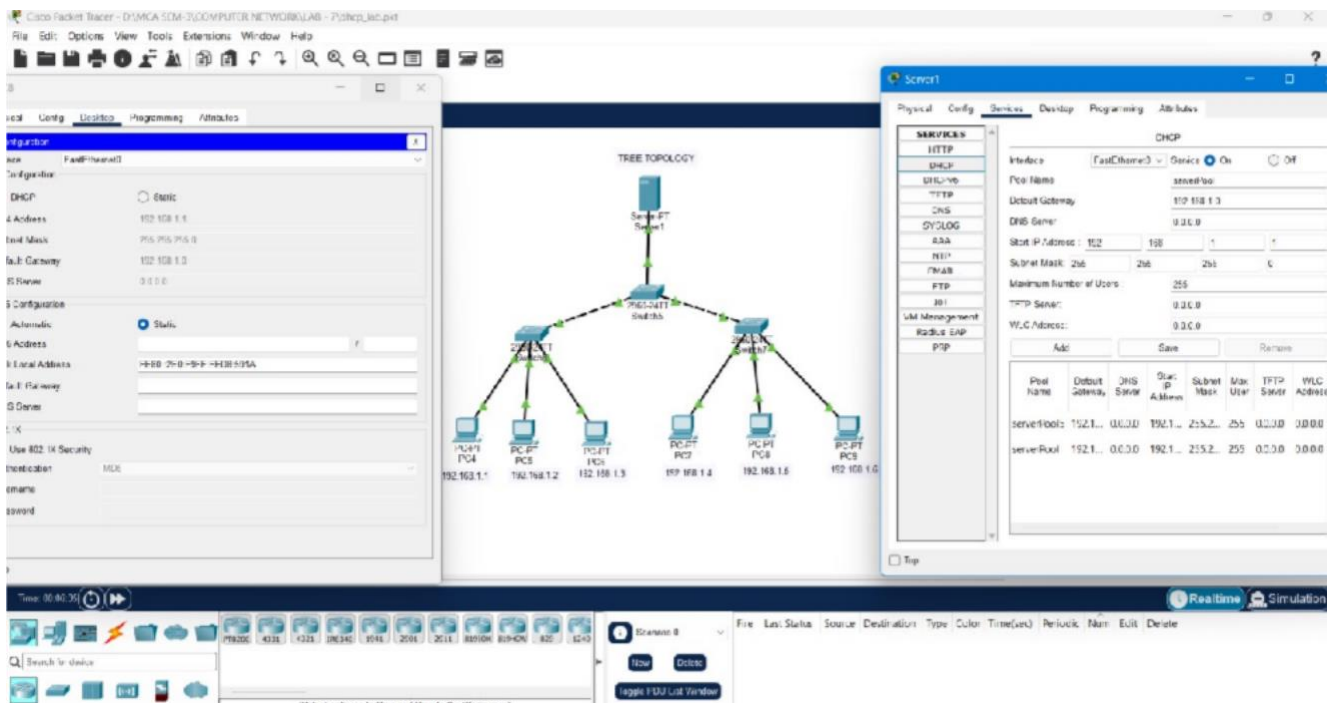
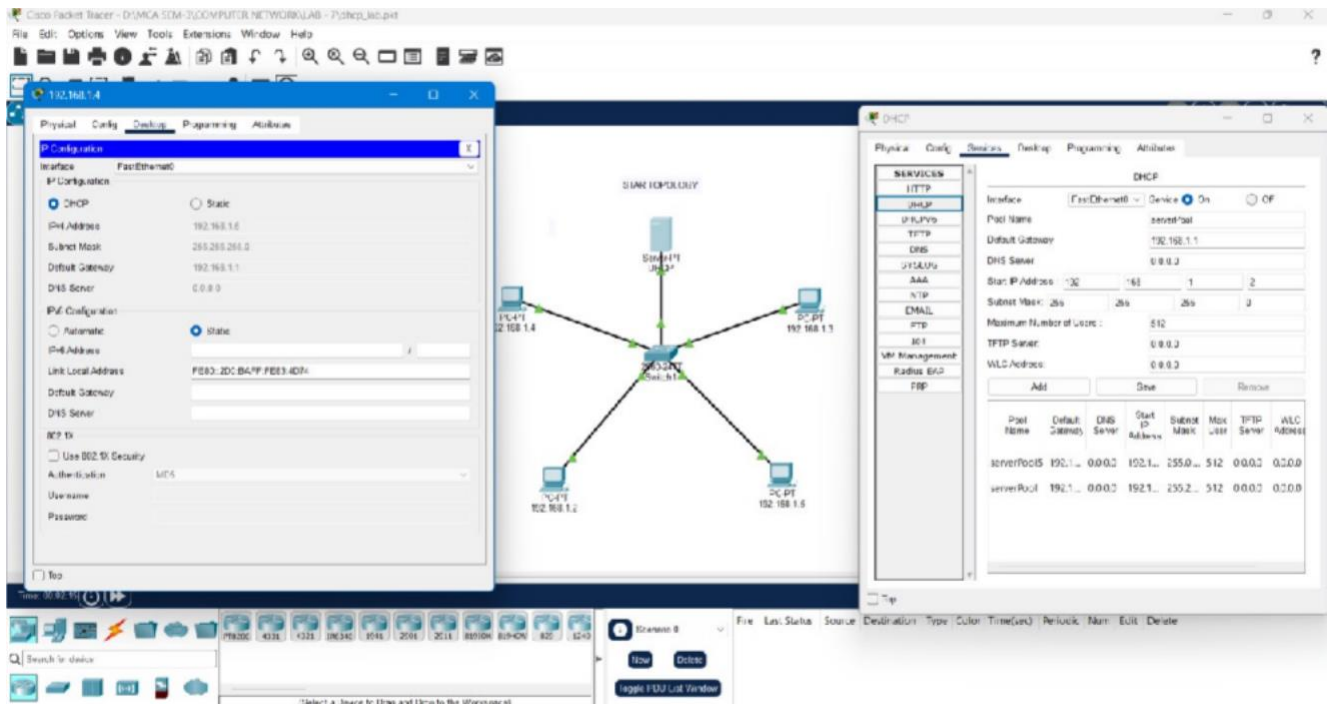
(Attach screenshot 1. Showing DHCP database 2. Assignment of IP address thru DHCP in PC).

The screenshot displays the Cisco Packet Tracer interface. The central workspace shows a Mesh Topology with four PCs (PC0, PC1, PC2, PC3) connected to a central switch (Switch1). The left pane shows the configuration for PC0, with the DHCP service enabled. The right pane shows the DHCP configuration for the switch, with the DHCP service enabled and the IP address 192.168.1.1 assigned to the switch's interface. The DHCP database is visible at the bottom right, showing the assignment of IP addresses to the PCs.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max. Users	TFTP Server	VLC Address
serverPool	192.168.1.1	0.0.0.0	192.168.1.2	255.255.255.0	254	0.0.0.0	0.0.0.0
serverPool	192.168.1.1	0.0.0.0	192.168.1.3	255.255.255.0	254	0.0.0.0	0.0.0.0



Date: 27/08/2026





Date: 10/09/2026

Lab Practical #07:

Study the application layer protocol FTP.

1. Implement the application layer protocol FTP. Also check connectivity between them using ping command or PDU utility.

Instructions:

1. Protocol-wise configuration setup screenshot.
2. Mention IP address of each pc as label.

Implementation:

FTP :

Create a network using 6 pc 1 switch 1 server where 2pc groups in Students , Admin and TA respectively

-configure ftp service through server and then create a file admin.txt through admin pc and share it over the network

-provide restricted the access rights as follows :

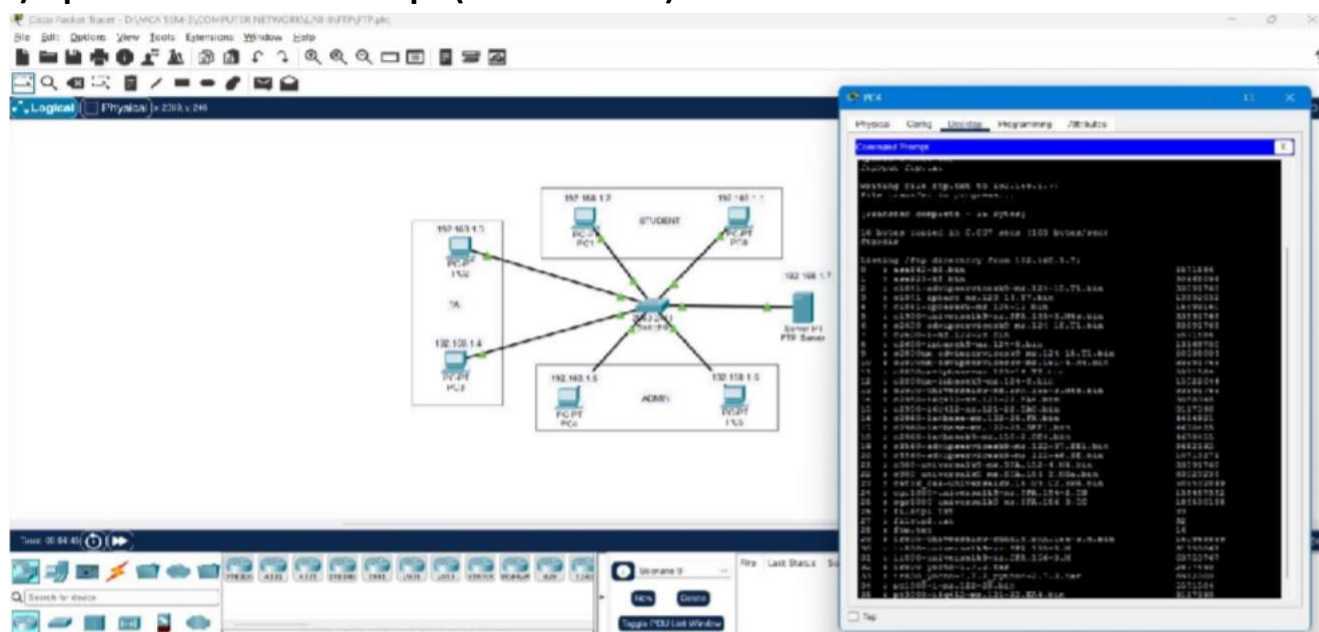
Student : List, Read

TA : Read , List , Rename

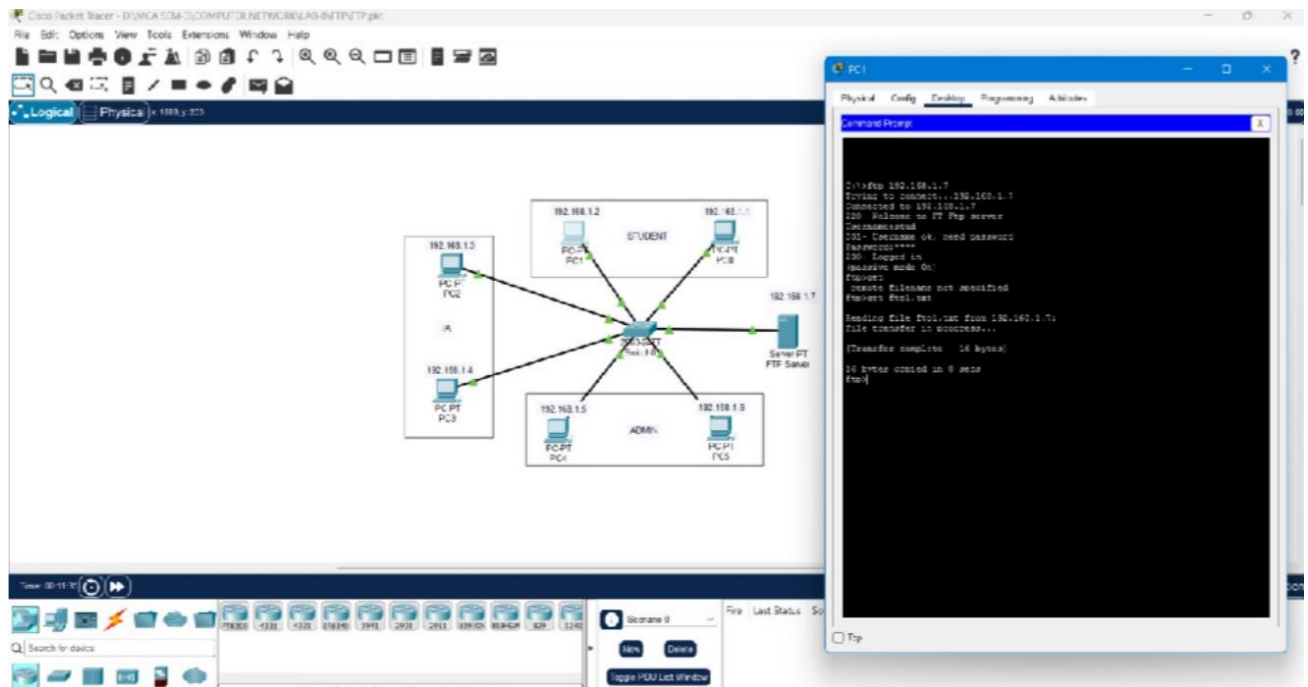
Admin: Read , List, Write , Rename , Delete

Task:

1) Upload file from Admin pc (1-Screenshot)

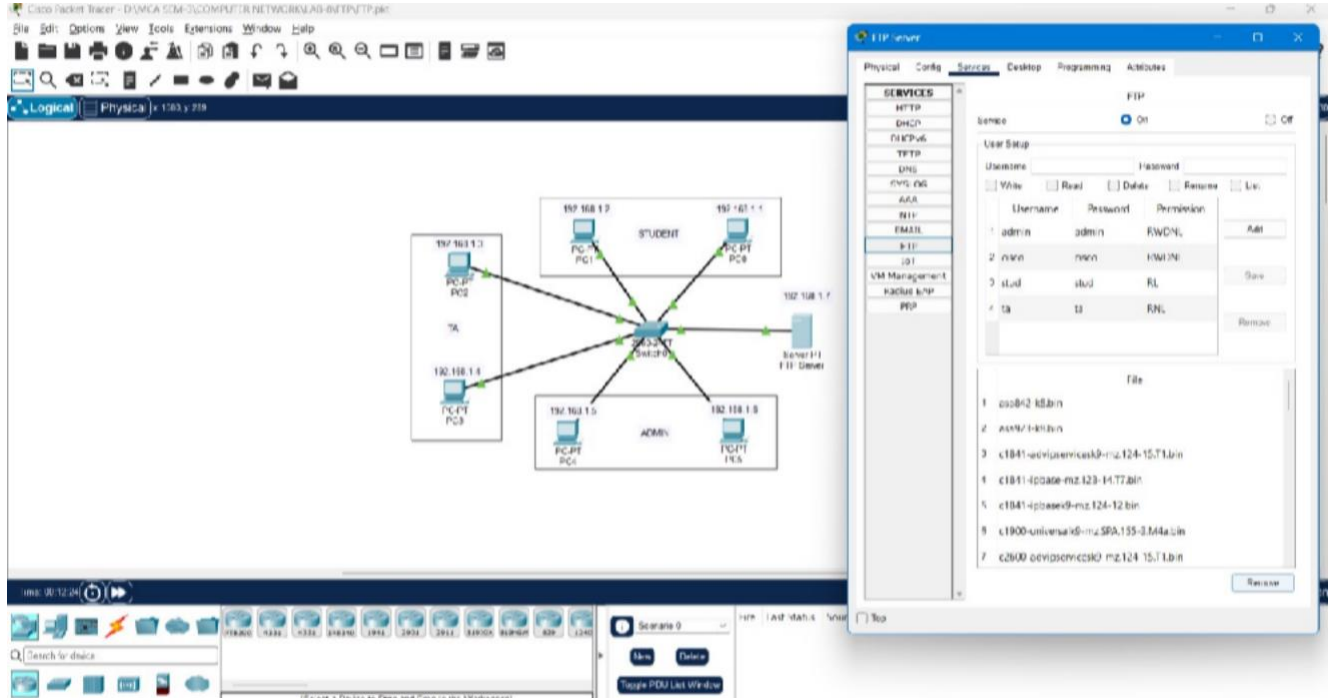


2) Rename admin.txt from TA pc (1-screenshot)



Date: 10/09/2026

4) Take one screenshot showing user database from FTP server



The screenshot displays a Cisco Packet Tracer workspace with a network topology. A central router (R1) is connected to several PCs and a server. The PCs are labeled with IP addresses: 192.168.1.3, 192.168.1.4, 192.168.1.5, 192.168.1.6, 192.168.1.7, and 192.168.1.8. The server is labeled 'Server1' with IP 192.168.1.9. The router is labeled 'R1' with IP 192.168.1.1. The topology is titled 'STUDENT'.

Overlaid on the right is the 'FTP Server' configuration window. The 'Services' tab is active, showing the 'FTP' service is enabled. The 'User Setup' section displays a table of users:

Username	Password	Permission
admin	admin	RWCDNL
anon	anon	RWCDNL
stud	stud	RL
ta	ta	RNL

The 'File' section lists the following files:

- 1. cs0842.k8.bin
- 2. cs0842.k8.bin
- 3. c1841-adviseemcsk8-mz.124-15.7.bin
- 4. c1841-adviseemcsk8-mz.124-15.7.bin
- 5. c1841-adviseemcsk8-mz.124-15.7.bin
- 6. c1900-universalk9-mz.SPA.155-3.M4a.bin
- 7. c2800-adviseemcsk8-mz.124-15.7.bin

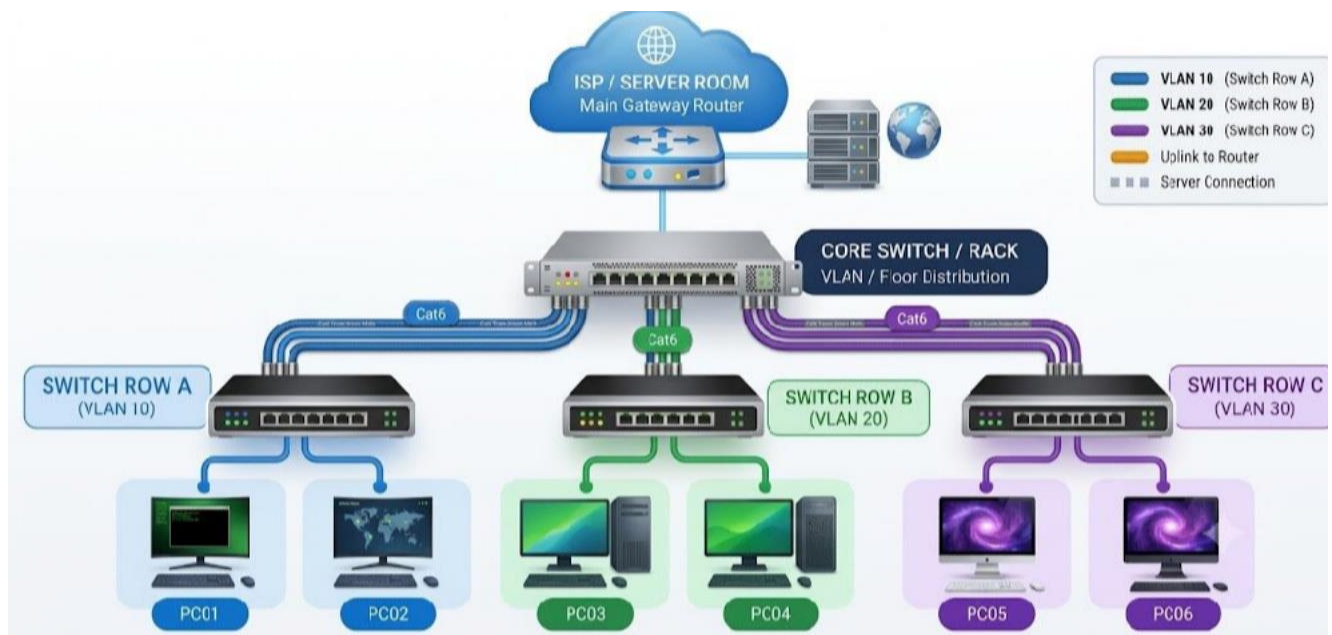
Date: 10/09/2026

Lab Practical #8:

Study & Survey of Institute organization network infrastructure.

Practical Assignment #8:

1. Identify type of network in your institute. Draw a design of network in your institute (Any Lab/Floor/Building).



**Date: 10/09/2026****2. List how many network devices and types of cable used and give its details.**

Category	Device / Cable Name	Standard Quantity (Per Floor / Lab Unit)	Details & Purpose
Network Cable	Cat6 UTP Cable	~30–60 Cables	Connects desktop PCs to the wall-mounted network switch inside the lab. Supports data transfer speeds up to 1 Gbps.
Network Cable	Single-Mode Fiber Optic Cable	1 Main Cable Line	Acts as the backbone link connecting the main Central Server Room to the departmental floor switch. Handles long distances without signal loss.
Network Cable	RJ-45 Connectors	60+ Units	Terminated on both ends of Cat6 ethernet cables to plug directly into PC Network Interface Cards (NICs) and switch ports.
Network Device	24/48-Port Managed Access Switch	1–2 Switches	Installed inside the computer lab rack to connect all local student PCs into a single local area network (LAN).
Network Device	Layer 3 Core Switch	1 Central Unit	Located in the main campus server room. Manages VLAN routing and traffic flow between different academic departments.
Network Device	Hardware Firewall (Sophos/Fortinet)	1 Central Unit	Sits between the Internet line and internal network to secure campus data, manage user authentication, and restrict unwanted traffic.
Network Device	Wi-Fi Access Points (Ubiquiti/Cisco)	2–4 Points	Mounted on corridor ceilings and lab areas to provide wireless internet access for mobile devices and laptops.



Date: 17/09/2026

Lab Practical #09:

Study Client-Server Socket programming - TCP & UDP

Practical Assignment #09:

- 1. Write a C/Java code for TCP Server-Client Socket Programming.**
- 2. Write a C/Java code for UDP Server-Client Socket Programming.**

1. For TCP Server-Client:

TCP Server Program:

```
import java.io.*;
import java.net.*;

public class TCPServer {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(5000);

            System.out.println("TCP Server started...");
            System.out.println("Server IP: 192.168.1.105");
            System.out.println("Waiting for client...");

            Socket socket = serverSocket.accept();

            System.out.println("Client connected!");

            BufferedReader input = new BufferedReader(new InputStreamReader(socket.getInputStream()));
            PrintWriter output = new PrintWriter(socket.getOutputStream(), true);

            String message = input.readLine();

            System.out.println("Client: " + message);

            output.println("Hello Client! Message received by Server.");

            socket.close();
            serverSocket.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```



Date: 17/09/2026

TCP Client Program:

```
import java.io.*;
import java.net.*;

public class TCPClient {
    public static void main(String[] args) {
        try {
            String serverIP = "192.168.1.105";

            Socket socket = new Socket(serverIP, 5000);

            System.out.println("Connected to TCP Server!");

            PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
            output.println("Hello Server!");

            BufferedReader input = new BufferedReader(new InputStreamReader(socket.getInputStream()));

            String response = input.readLine();

            System.out.println("Server: " + response);

            socket.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

TCP Server Output:

```
TCP Server started...
Server IP: 192.168.1.105
Waiting for client...
Client connected!
Client: Hello Server!
```

TCP Client Output:

```
Connected to TCP Server!
Server: Hello Client! Message received by Server.
```



Date: 17/09/2026

2. For UDP Server-Client:

UDP Server Program:

```
import java.net.*;

public class UDPServer {
    public static void main(String[] args) {
        try {
            DatagramSocket socket = new DatagramSocket(6000);

            System.out.println("UDP Server started...");
            System.out.println("Server IP: 192.168.1.105");
            System.out.println("Waiting for client...");

            byte[] receiveData = new byte[1024];

            DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);

            socket.receive(receivePacket);

            String message = new String(receivePacket.getData(), 0, receivePacket.getLength());

            System.out.println("Client: " + message);

            String response = "Hello Client! UDP message received.";

            byte[] sendData = response.getBytes();

            DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length,
            receivePacket.getAddress(), receivePacket.getPort());

            socket.send(sendPacket);

            socket.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

UDP Server Output:

```
UDP Server started...
Server IP: 192.168.1.105
Waiting for client...
Client: Hello UDP Server!
```



Date: 17/09/2026

UDP Client Program:

```
import java.net.*;

public class UDPClient {
    public static void main(String[] args) {
        try {
            DatagramSocket socket = new DatagramSocket();

            String serverIP = "192.168.1.105";

            InetAddress address = InetAddress.getByName(serverIP);

            String message = "Hello UDP Server!";

            byte[] sendData = message.getBytes();

            DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length, address, 6000);

            socket.send(sendPacket);

            System.out.println("Message sent to UDP Server!");

            byte[] receiveData = new byte[1024];

            DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);

            socket.receive(receivePacket);

            String response = new String(receivePacket.getData(), 0, receivePacket.getLength());

            System.out.println("Server: " + response);

            socket.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

UDP Client Output

Message sent to UDP Server!
Server: Hello Client! UDP message received.



Date: 17/09/26

Lab Practical #10:

Study Packet capture and header analysis by Wireshark (HTTP, ICMP, DNS, TCP, UDP, ip addr etc.)

Practical Assignment #10:

1. Explain usage of Wireshark tool.

Ans.

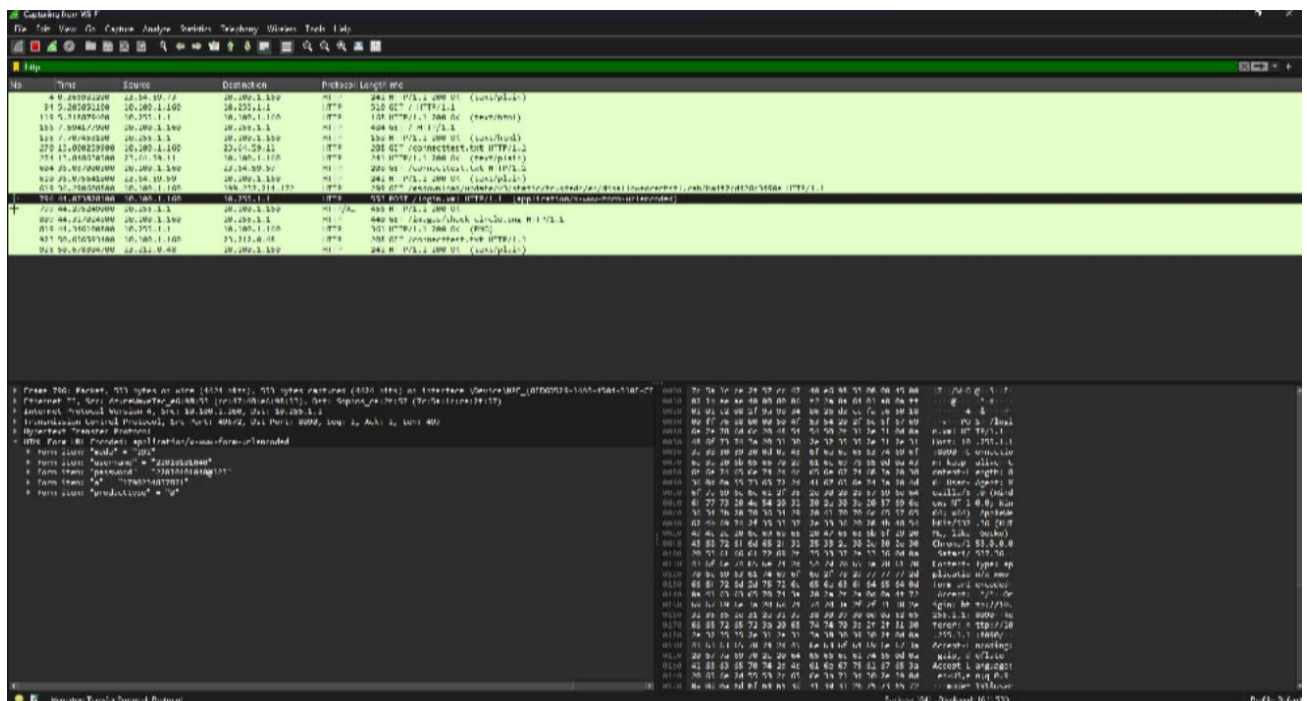
Wireshark is a network packet analyzer used to capture, monitor, and analyze network traffic.

Uses:

- Captures network packets.
- Analyzes protocols like TCP, UDP, HTTP, and DNS.
- Helps troubleshoot network problems.
- Uses filters to find specific packets.
- Helps detect unusual or suspicious network activity.

2. Packet capture and header analysis by Wireshark (1.HTTP, 2.ICMP, 3.DNS, 4.TCP, 5.UDP, 6.ip addr).

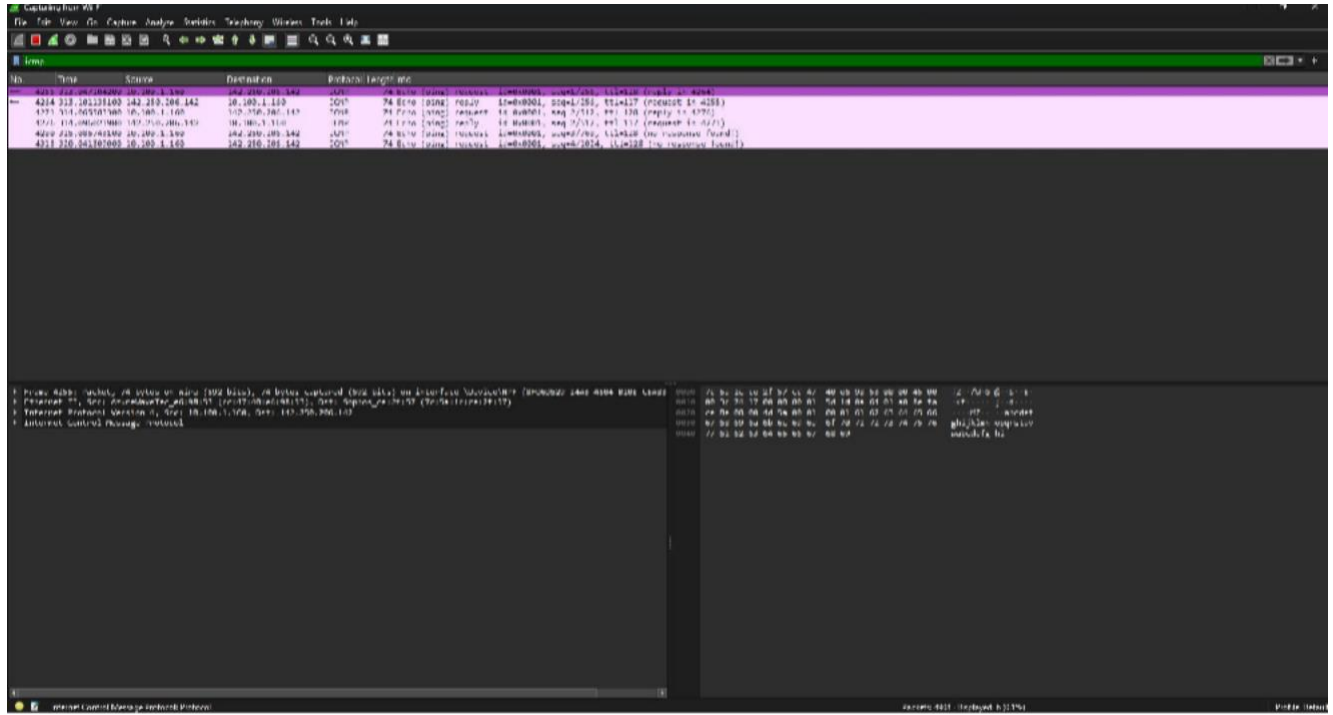
1. HTTP



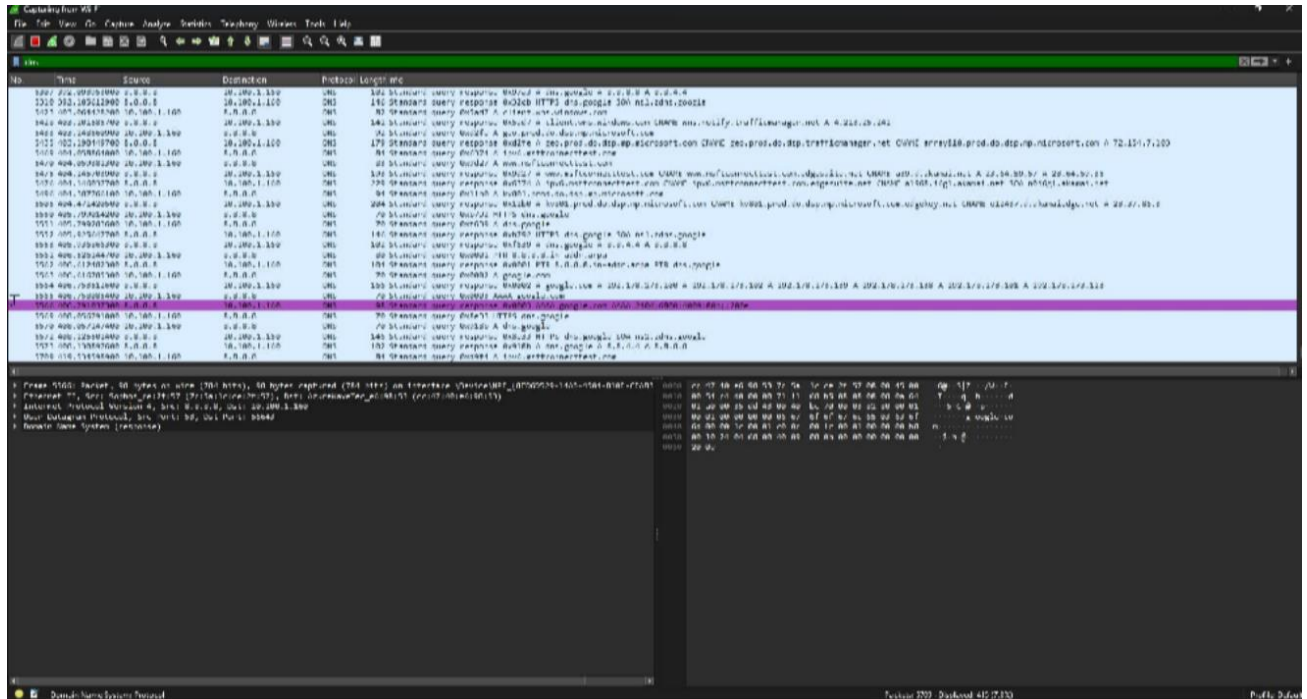


Date: 17/09/24

2. ICMP



3. DNS



Date: 17/09/26

4. TCP

[illegible]

5. UDP

[illegible]



6. Ip address

52

Date: 26/09/26**Lab Practical #12:**

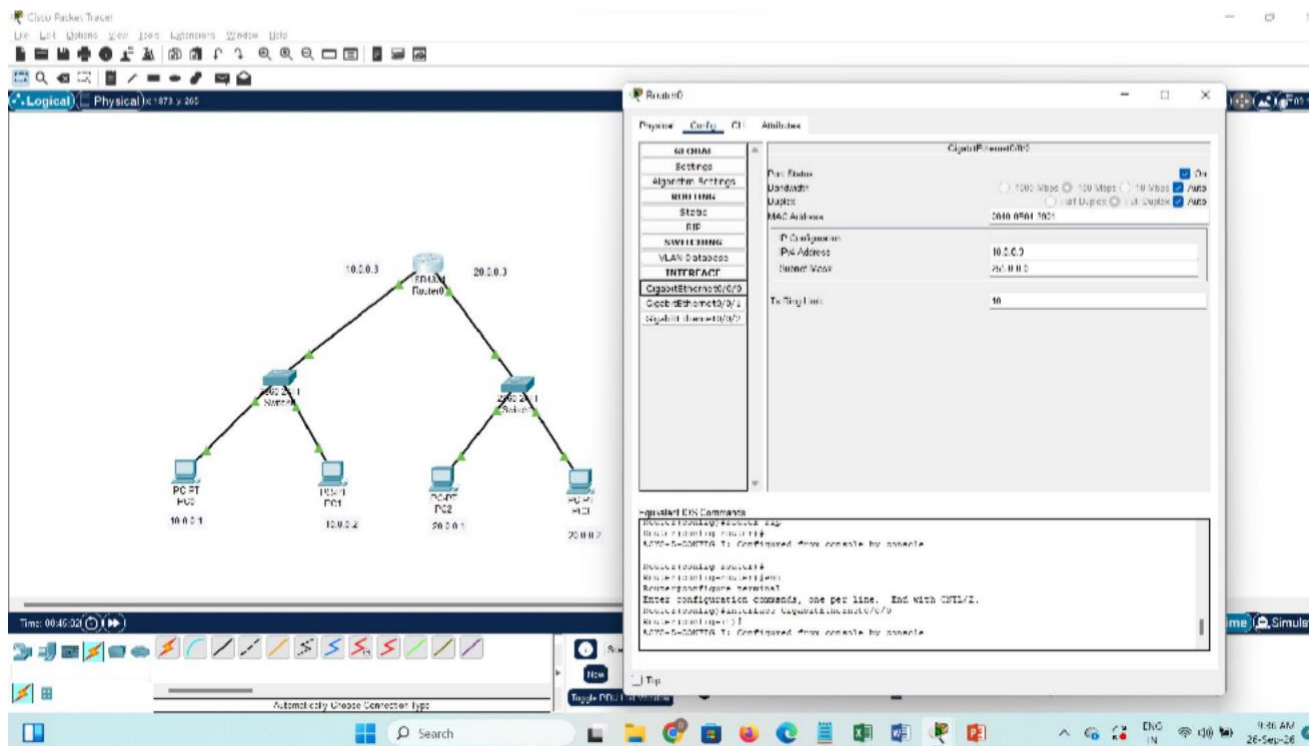
Study the concept of routing using packet tracer. (Dynamic Routing)

Practical Assignment #12:

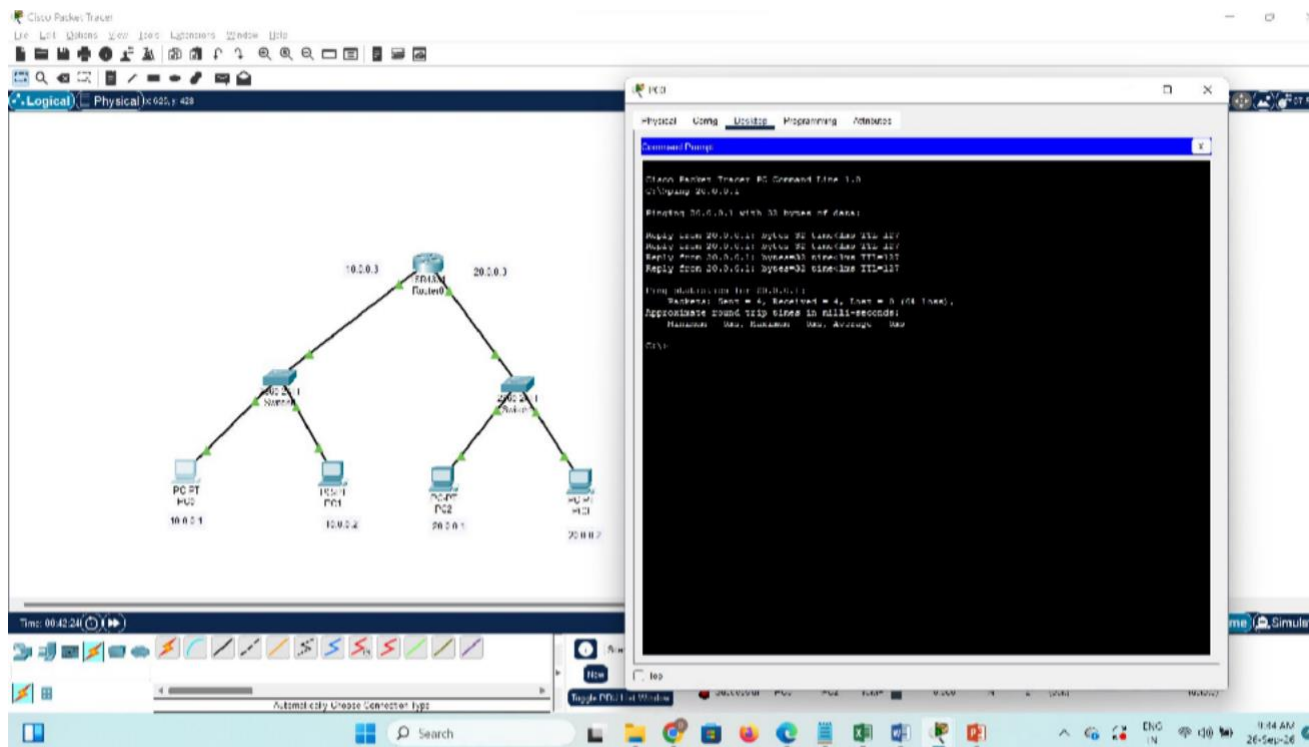
1. Connect the two different networks based on the calculated IP addresses and subnet using a packet tracer.
2. Connect the three different networks based on the calculated IP addresses and subnet using a packet tracer.

Instructions:

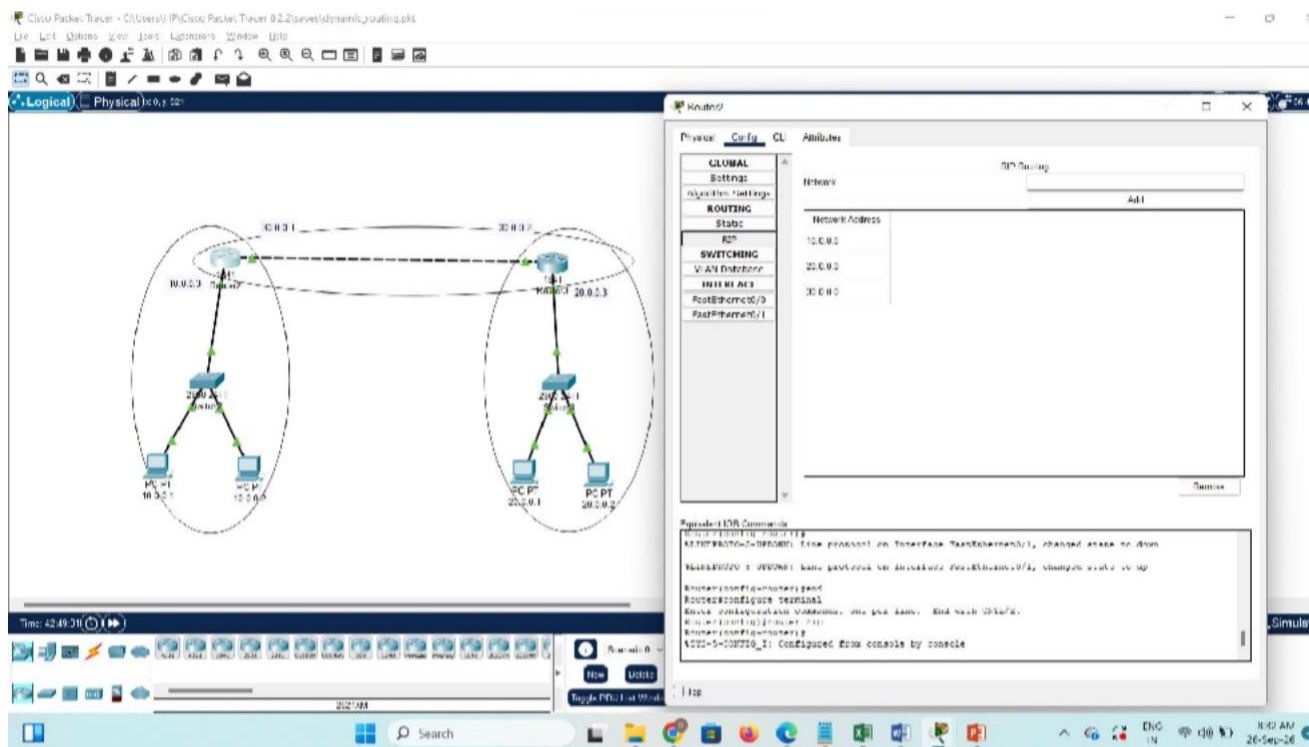
1. RIP routing screenshot with routing table. (Take two or more different networks)
2. Mention IP address of each node and network ID of each network as label.
3. Ping command / Packet transfer screenshot between two different network nodes.

1. Two different networks

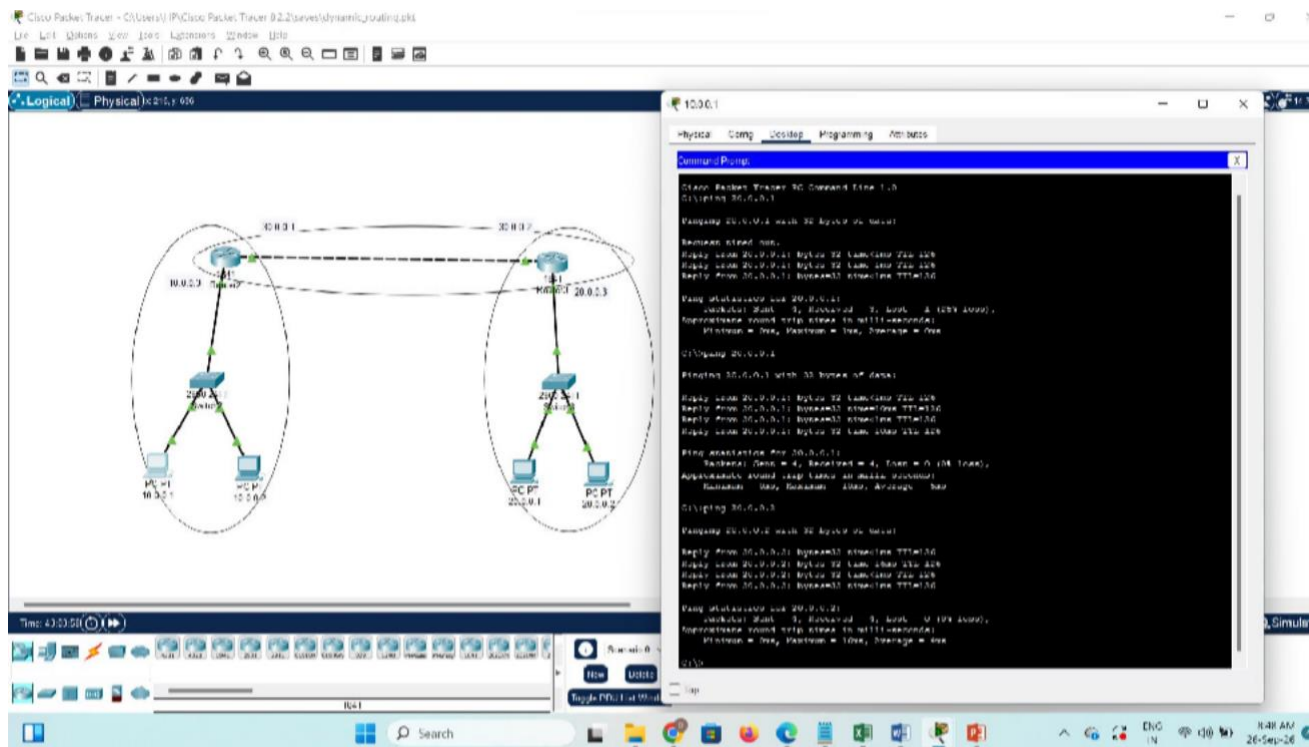
Date: 26/09/26



2. Three different networks



Date: 26/09/26



Date: 26/09/26**Lab Practical #13:**

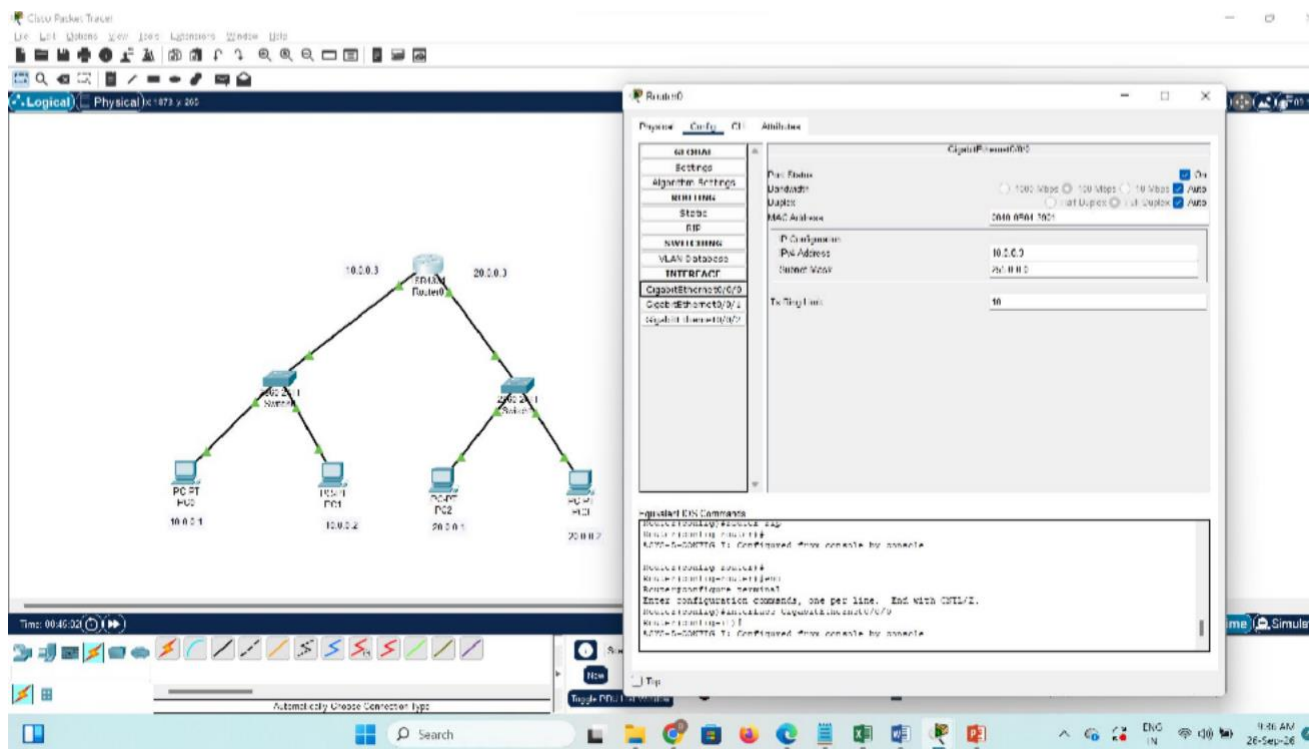
Study the concept of routing using packet tracer. (Static Routing)

Practical Assignment #13:

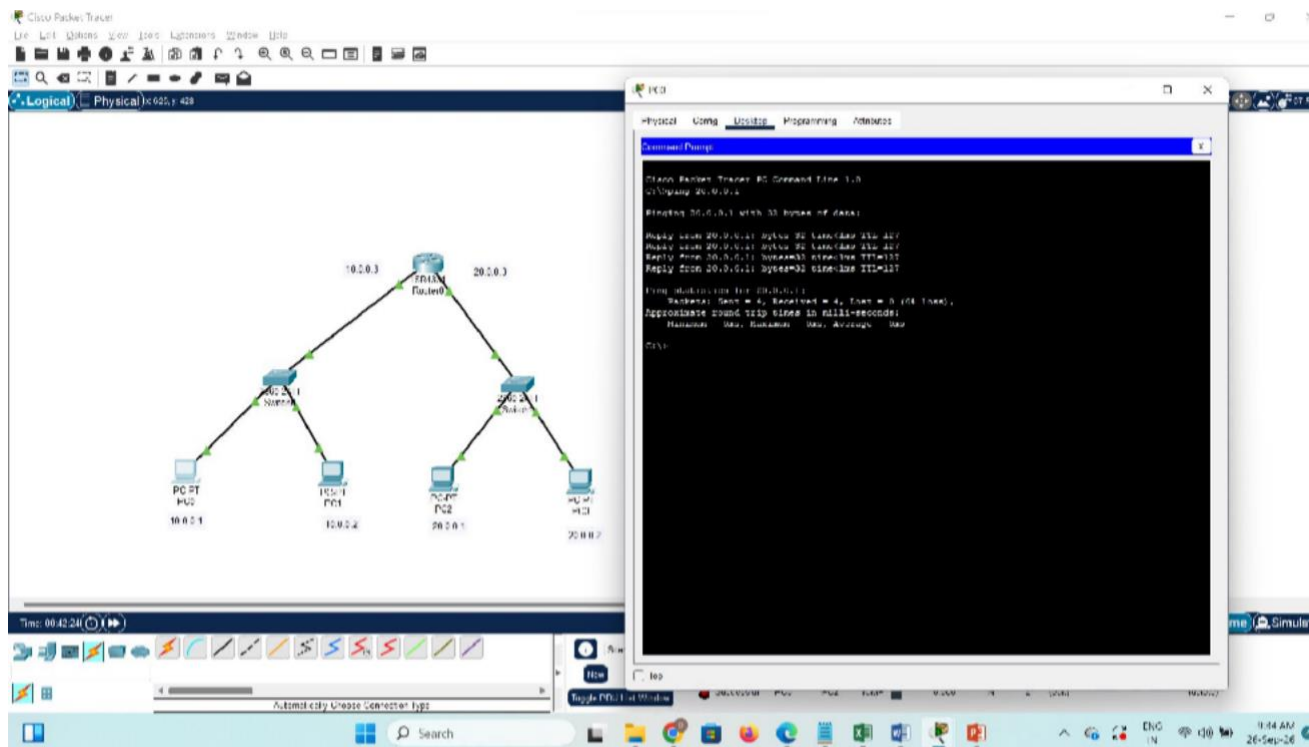
1. Connect the two different networks based on the calculated IP addresses and subnet using a packet tracer.
2. Connect the three different networks based on the calculated IP addresses and subnet using a packet tracer.

Instructions:

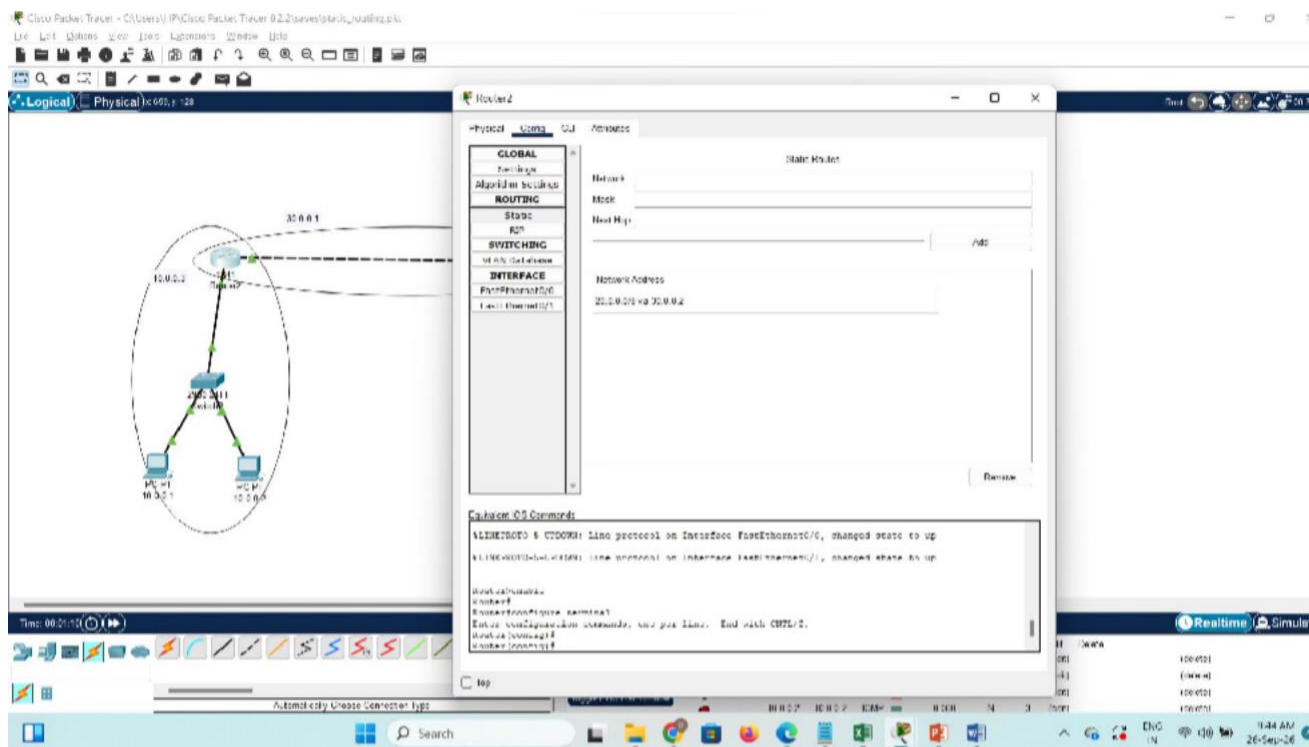
1. Static routing screenshot with routing table. (Take two or more different networks)
2. Mention IP address of each node and network ID of each network as label.
3. Ping command / Packet transfer screenshot between two different network nodes.

1. Two different networks

Date: 26/09/26



2. Three different networks



Date: 26/09/26

